Practice 1 — Set Theory

Solutions to Selected Problems

Problems 2, 7, 11, 16, 20, 54 (3, 11), 58, 60

Discrete Mathematics — QYSU

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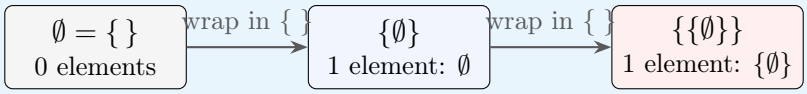
1 Problem 2 — Membership with $\emptyset$ and $\{\emptyset\}$

Theory Recap: Membership ($\in$) vs. Subset ($\subseteq$)

These two relations are fundamentally different:

- $a \in S$ means "a is an **element** of S."
- $A \subseteq S$ means "every element of A is also in S."

The empty set $\emptyset$ has no elements. The set $\{\emptyset\}$ is **not** empty—it has one element, namely $\emptyset$ itself. To decide "$x \in S$?" list the elements of S and check whether x is one of them.



Each wrapping step creates a new set whose sole element is the previous set.

Problem Statement

Which of the following statements are true?

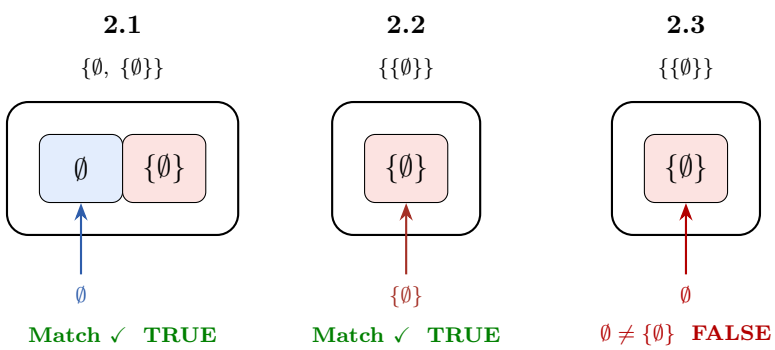
2.1. $\emptyset \in \{\emptyset,\ \{\emptyset\}\}$

2.2. $\{\emptyset\} \in \{\{\emptyset\}\}$

2.3. $\emptyset \in \{\{\emptyset\}\}$

Solution

The strategy is always the same: *list the elements* of the outer set, then check whether the object on the left of $\in$ matches any of them.



2.1. $\emptyset \in \{\emptyset,\ \{\emptyset\}\}$ — **TRUE.** The set has two elements: $\emptyset$ and $\{\emptyset\}$. The object $\emptyset$ is explicitly one of them.

2.2. $\{\emptyset\} \in \{\{\emptyset\}\}$ — **TRUE.** The set $\{\{\emptyset\}\}$ has exactly one element: $\{\emptyset\}$. The object on the left is $\{\emptyset\}$, which matches.

2.3. $\emptyset \in \{\{\emptyset\}\}$ — **FALSE.** The only element is $\{\emptyset\}$. We need $\emptyset = \{\emptyset\}$, but $|\emptyset| = 0$ while $|\{\emptyset\}| = 1$, so they are not equal.

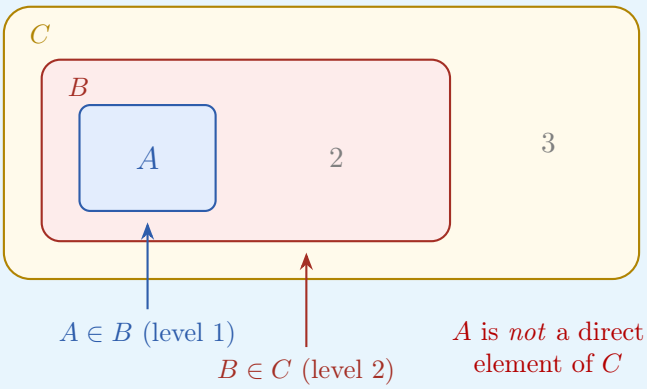
2.1 — True 2.2 — True 2.3 — False

2 Problem 7 — $\in$ is Not Transitive

Theory Recap: $\in$ is Not Transitive

$\subseteq$ **is** transitive: $A \subseteq B \wedge B \subseteq C \implies A \subseteq C$.
$\in$ is **not**: $A \in B \wedge B \in C \not\Longrightarrow A \in C$.

Why? $\in$ crosses one "level" of nesting. Two $\in$-steps cross *two* levels, so A is buried inside C but is not a direct element of C.



Problem Statement

Give examples of sets A, B, C such that $A \in B$, $B \in C$, but $A \notin C$.

Solution

Let

$$A = 1, \qquad B = \{1, 2\}, \qquad C = \big\{\{1, 2\},\ 3\big\}.$$

Condition	Check	Result
$A \in B$	Is $1 \in \{1, 2\}$?	Yes ✓
$B \in C$	Is $\{1, 2\} \in \big\{\{1, 2\},\ 3\big\}$?	Yes ✓
$A \notin C$	Is $1 \in \big\{\{1, 2\},\ 3\big\}$? The elements are $\{1, 2\}$ and 3.	$1 \notin C$ ✓

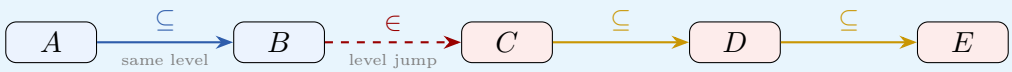
Key insight. The number 1 is an element of B, and B is an element of C, but the elements of C are $\{1, 2\}$ (a set) and 3 (a number). The number 1 is neither of these, so $1 \notin C$.

3 Problem 11 — Mixing $\in$ and $\subseteq$

Theory Recap: Mixing $\in$ and $\subseteq$

The two relations operate at different "levels":

- $A \subseteq B$: both sit at the same level—every element of A is also an element of B.
- $B \in C$: B itself becomes an object inside C—a jump *up* one level.



Problem Statement

Give examples of sets A, B, C, D, E satisfying:

$$A \subseteq B, \quad B \in C, \quad C \subseteq D, \quad D \subseteq E.$$

Solution

$$A = \{1\}, \quad B = \{1, 2\}, \quad C = \big\{\{1, 2\},\ 3\big\}, \quad D = \big\{\{1, 2\},\ 3,\ 4\big\}, \quad E = \big\{\{1, 2\},\ 3,\ 4,\ 5\big\}$$

Condition	Type	Verification
$A \subseteq B$	subset	Every element of $\{1\}$ is in $\{1, 2\}$: $1 \in \{1, 2\}$ ✓
$B \in C$	membership	$\{1, 2\}$ is listed as an element of $\big\{\{1, 2\}, 3\big\}$ ✓
$C \subseteq D$	subset	Elements $\{1, 2\}$ and 3 of C are both in D ✓
$D \subseteq E$	subset	Elements $\{1, 2\}$, 3, 4 of D are all in E ✓

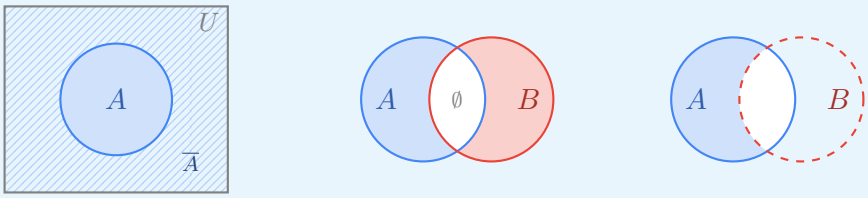
Why this works. $\subseteq$ preserves the "level": A and B contain the same kind of objects (numbers). The $\in$-step promotes B to an object inside C, after which C, D, E all live at the higher level (their elements include sets like $\{1, 2\}$).

4 Problem 16 — Describing Sets Under Operations

Theory Recap: Set Operations

For subsets of a universal set U:

Operation	Symbol	Definition
Complement	$\overline{A}$	$\{x \in U \mid x \notin A\}$
Set difference	$A \setminus B$	$\{x \in A \mid x \notin B\}$
Symmetric diff.	$A \triangle B$	$(A \setminus B) \cup (B \setminus A)$ = elements in exactly one of A, B



Complement $\quad A \triangle B \quad A \setminus B$

Problem Statement

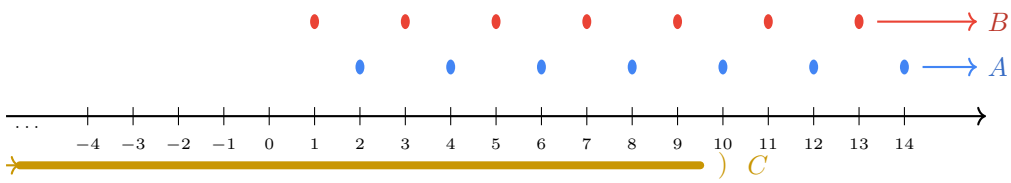
Let $\mathbb{Z}$ be the universal set. Define:

$$A = \{x \in \mathbb{Z} \mid \exists\, y \in \mathbb{Z}^+,\ x = 2y\} \qquad \text{(positive even integers: } 2, 4, 6, 8, \ldots)$$
$$B = \{x \in \mathbb{Z} \mid \exists\, y \in \mathbb{Z}^+,\ x = 2y - 1\} \qquad \text{(positive odd integers: } 1, 3, 5, 7, \ldots)$$
$$C = \{x \in \mathbb{Z} \mid x < 10\}$$

Describe: $\overline{A}$, $A \triangle B$, $\overline{C}$, $A \setminus C$, $C \setminus (A \triangle B)$.

Solution

Visualising the sets on the number line.



(a) $\overline{A}$**:** Everything in $\mathbb{Z}$ that is *not* a positive even integer:

$$\boxed{\overline{A} = \{\ldots, -2, -1, 0\} \cup \{1, 3, 5, 7, \ldots\} = \{x \in \mathbb{Z} \mid x \leq 0 \text{ or } x \text{ is odd}\}}$$

(b) $A \triangle B$**:** Since $A \cap B = \emptyset$ (no integer is both positive-even and positive-odd):

$$A \triangle B = A \cup B = \{1, 2, 3, 4, \ldots\} \qquad \Longrightarrow \qquad \boxed{A \triangle B = \mathbb{Z}^+}$$

(c) $\overline{C}$**:** All integers ≥ 10: $\boxed{\overline{C} = \{10, 11, 12, \ldots\}}$

(d) $A \setminus C$**:** Positive evens that are ≥ 10: $\boxed{A \setminus C = \{10, 12, 14, \ldots\} = \{2k \mid k \geq 5\}}$

(e) $C \setminus (A \triangle B)$**:** Remove all positive integers ($= A \triangle B$) from C:

$$\boxed{C \setminus (A \triangle B) = \{\ldots, -2, -1, 0\} = \{x \in \mathbb{Z} \mid x \leq 0\}}$$

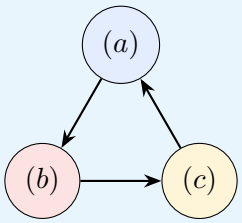
5 Problem 20 — Equivalent Characterizations of $\subseteq$

Theory Recap: Cyclic-Implication Proof Strategy

To show three conditions (a), (b), (c) are equivalent, it suffices to prove a **cycle of implications**:

$$(a) \implies (b) \implies (c) \implies (a).$$

This is more efficient than proving all six pairwise implications directly.

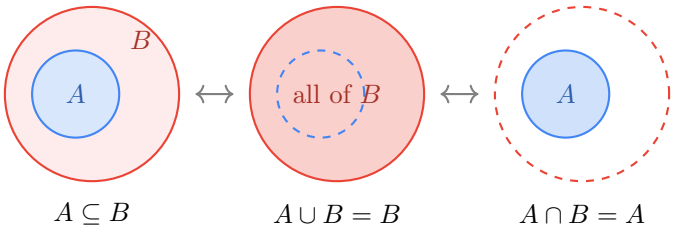


Problem Statement

Prove that for any subsets A, B of universal set U, the conditions in each group are equivalent:

1. $A \subseteq B \iff A \cup B = B \iff A \cap B = A$
2. $A \cap B = \emptyset \iff A \subseteq \overline{B} \iff B \subseteq \overline{A}$
3. $A \cup B = U \iff \overline{A} \subseteq B \iff \overline{B} \subseteq A$

Part 1: $A \subseteq B \iff A \cup B = B \iff A \cap B = A$



$(a) \Rightarrow (b)$**:** $A \subseteq B \implies A \cup B = B$.

$(\supseteq)$ $B \subseteq A \cup B$ always.

$(\subseteq)$ Let $x \in A \cup B$. Then $x \in A$ or $x \in B$. If $x \in A$, then $A \subseteq B$ gives $x \in B$. Either way, $x \in B$. $\square$

$(b) \Rightarrow (c)$**:** $A \cup B = B \implies A \cap B = A$.

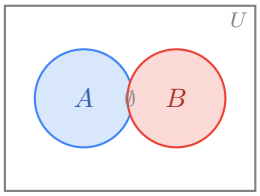
$(\subseteq)$ $A \cap B \subseteq A$ always.

$(\supseteq)$ Let $x \in A$. Then $x \in A \cup B = B$, so $x \in B$. Hence $x \in A \cap B$. $\square$

$(c) \Rightarrow (a)$**:** $A \cap B = A \implies A \subseteq B$.

Let $x \in A$. Then $x \in A = A \cap B$, so $x \in B$. $\square$

Part 2: $A \cap B = \emptyset \iff A \subseteq \overline{B} \iff B \subseteq \overline{A}$



Disjoint: $A \cap B = \emptyset$

$(a) \Rightarrow (b)$**:** $A \cap B = \emptyset \implies A \subseteq \overline{B}$.

Let $x \in A$. If $x \in B$, then $x \in A \cap B = \emptyset$—contradiction. So $x \notin B$, i.e. $x \in \overline{B}$. $\square$

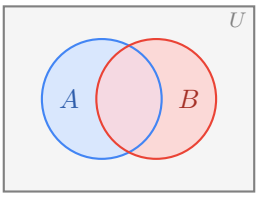
$(b) \Rightarrow (c)$**:** $A \subseteq \overline{B} \implies B \subseteq \overline{A}$.

Let $x \in B$. If $x \in A$, then $x \in \overline{B}$ (since $A \subseteq \overline{B}$), so $x \notin B$—contradiction. Hence $x \notin A$, i.e. $x \in \overline{A}$. $\square$

$(c) \Rightarrow (a)$**:** $B \subseteq \overline{A} \implies A \cap B = \emptyset$.

Suppose $x \in A \cap B$. Then $x \in B \subseteq \overline{A}$, so $x \notin A$—contradicting $x \in A \cap B$. $\square$

Part 3: $A \cup B = U \iff \overline{A} \subseteq B \iff \overline{B} \subseteq A$



Covering: $A \cup B = U$

$(a) \Rightarrow (b)$**:** $A \cup B = U \implies \overline{A} \subseteq B$.

Let $x \in \overline{A}$, so $x \notin A$. Since $x \in U = A \cup B$ and $x \notin A$, we must have $x \in B$. $\square$

$(b) \Rightarrow (c)$**:** $\overline{A} \subseteq B \implies \overline{B} \subseteq A$.

Let $x \in \overline{B}$, so $x \notin B$. If $x \notin A$, then $x \in \overline{A} \subseteq B$, giving $x \in B$—contradiction. So $x \in A$. $\square$

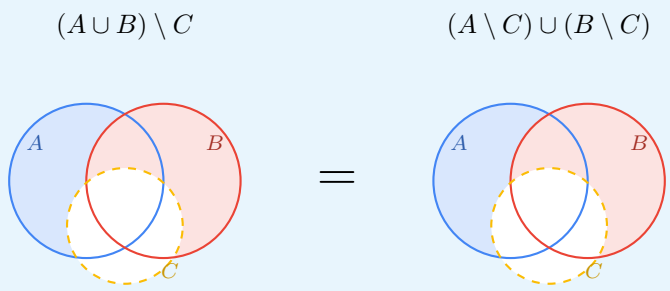
$(c) \Rightarrow (a)$**:** $\overline{B} \subseteq A \implies A \cup B = U$.

Let $x \in U$. Either $x \in B$ (then $x \in A \cup B$) or $x \notin B$ (then $x \in \overline{B} \subseteq A$, so $x \in A \cup B$). Since also $A \cup B \subseteq U$, we get $A \cup B = U$. $\square$

6 Problem 54.3 — $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$

Theory Recap: Set Difference Distributes Over Union

Mental model: "$X \setminus C$" means "start with X, **remove** everything in C."
Removing C from a union is the same as removing C from each piece separately, then taking the union of the remainders.



Both diagrams shade exactly the same region—the parts of A and B that fall outside C.

Problem Statement

Prove: $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

Solution

Let x be arbitrary:

$$\begin{aligned}
x \in (A \cup B) \setminus C &\iff x \in (A \cup B) \ \wedge \ x \notin C \\
&\iff (x \in A \ \vee \ x \in B) \ \wedge \ x \notin C \\
&\iff (x \in A \wedge x \notin C) \ \vee \ (x \in B \wedge x \notin C) \qquad (\wedge \text{ distributes over } \vee) \\
&\iff x \in (A \setminus C) \ \vee \ x \in (B \setminus C) \\
&\iff x \in (A \setminus C) \cup (B \setminus C).
\end{aligned}$$

Every step is $\iff$, so the sets are equal. $\square$

Logic correspondence. The key step uses the propositional law $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$, where $p = (x \notin C)$, $q = (x \in A)$, $r = (x \in B)$.

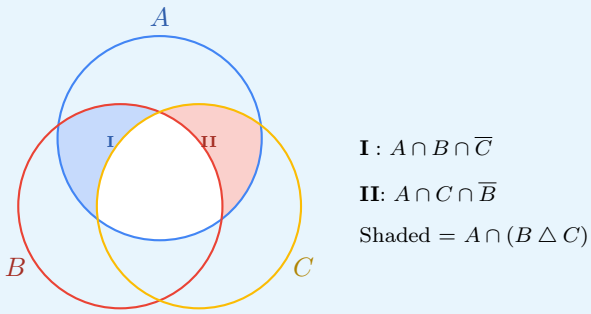
7 Problem 54.11 — $\cap$ Distributes Over $\triangle$

Theory Recap: Intersection Distributes Over Symmetric Difference

Recall: $X \triangle Y = \{x \mid x \text{ belongs to exactly one of } X, Y\}$.

The identity $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$ says:

> *"The part of A that sees exactly one of B, C"* equals *"exactly one of (A's overlap with B) and (A's overlap with C)."*



Algebraic analogy. Under the correspondence $\cap \leftrightarrow \cdot$ and $\triangle \leftrightarrow +$ (mod 2), the power set $(\mathcal{P}(U), \triangle, \cap)$ forms a commutative ring. This identity is the ring distributive law.

Problem Statement

Prove: $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$.

Solution

Let x be arbitrary. We expand both sides to the same logical form.

Left-hand side:

$$\begin{aligned} x \in A \cap (B \triangle C) &\iff x \in A \ \wedge \ x \in B \triangle C \\ &\iff x \in A \ \wedge \ \big[(x \in B \wedge x \notin C) \ \vee \ (x \notin B \wedge x \in C)\big] \\ &\iff \underbrace{(x \in A \wedge x \in B \wedge x \notin C)}_{\text{case 1}} \vee \underbrace{(x \in A \wedge x \notin B \wedge x \in C)}_{\text{case 2}} \qquad (\star) \end{aligned}$$

Right-hand side:

$$x \in (A \cap B) \triangle (A \cap C) \iff \big[x \in A \cap B \ \wedge \ x \notin A \cap C\big] \ \vee \ \big[x \notin A \cap B \ \wedge \ x \in A \cap C\big]$$

First disjunct: $x \in A \cap B$ and $x \notin A \cap C$.

- $x \in A \cap B \implies x \in A$ and $x \in B$.
- $x \notin A \cap C$ means $\neg(x \in A \wedge x \in C)$. Since $x \in A$, this forces $x \notin C$.
- Combined: $x \in A,\ x \in B,\ x \notin C$ (= case 1).

Second disjunct: $x \notin A \cap B$ and $x \in A \cap C$.

- $x \in A \cap C \implies x \in A$ and $x \in C$.

- $x \notin A \cap B$ means $\neg(x \in A \wedge x \in B)$. Since $x \in A$, this forces $x \notin B$.
- Combined: $x \in A,\ x \notin B,\ x \in C$ (= case 2).

Both sides reduce to $(\star)$, so:

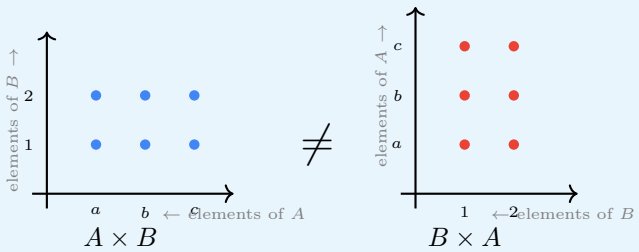
$$A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C). \qquad \blacksquare$$

8 Problem 58 — $A \times B = B \times A \implies A = B$

Theory Recap: Cartesian Product

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

Order in the pair matters: $(a, b) \neq (b, a)$ when $a \neq b$.



When $A \neq B$, the grids have different shapes, so $A \times B \neq B \times A$.

Problem Statement

Prove that $A \times B = B \times A$ implies $A = B$.

Note: More precisely, $A \times B = B \times A$ implies $A = B$, or $A = \emptyset$, or $B = \emptyset$.

Solution

Assume $A \times B = B \times A$ with $A \neq \emptyset$ and $B \neq \emptyset$. We show $A \subseteq B$ and $B \subseteq A$.

$A \subseteq B$: Let $a \in A$. Since $B \neq \emptyset$, pick some $b \in B$.

$$(a, b) \in A \times B = B \times A \quad \implies \quad \text{first component } a \in B.$$

Since a was arbitrary, $A \subseteq B$.

$B \subseteq A$: Let $b \in B$. Since $A \neq \emptyset$, pick some $a \in A$.

$$(b, a) \in B \times A = A \times B \quad \implies \quad \text{first component } b \in A.$$

Since b was arbitrary, $B \subseteq A$.

By antisymmetry of $\subseteq$: $A \subseteq B \ \wedge\ B \subseteq A \implies A = B$. $\square$

Why non-emptiness matters. If $A = \emptyset$ and $B = \{1\}$, then $A \times B = \emptyset = B \times A$, but $A \neq B$. We cannot "pick $b \in B$" if $B = \emptyset$, so the argument breaks down.

9 Problem 60 — $\times$ Distributes Over Set Operations

Theory Recap: $\times$ Distributes Over $\cup$, $\cap$, $\setminus$

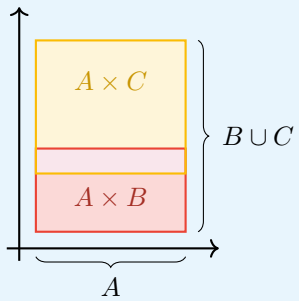
All three proofs follow the same three-step recipe:

(1) **Unpack:** $(x, y) \in A \times B \iff x \in A \wedge y \in B$.
(2) **Apply a propositional-logic law** ($\wedge/\vee$ distribution, or negation).
(3) **Repack** into set notation.

The operation in the second component "passes through" $\times$:

$$A \times (B \diamond C) = (A \times B) \diamond (A \times C) \qquad \text{for } \diamond \in \{\cup, \cap, \setminus\}.$$

Mirror identities $(B \diamond C) \times A$ work the same way on the *first* component. We prove the three non-mirror versions; the mirrors follow by swapping $x \leftrightarrow y$.



Problem Statement

Prove the following identities:

1. $A \times (B \cup C) = (A \times B) \cup (A \times C)$
3. $A \times (B \cap C) = (A \times B) \cap (A \times C)$
5. $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

Identity 1: $A \times (B \cup C) = (A \times B) \cup (A \times C)$

Let (x, y) be an arbitrary ordered pair:

$$\begin{aligned}
(x, y) \in A \times (B \cup C) &\iff x \in A \ \wedge \ y \in B \cup C \\
&\iff x \in A \ \wedge \ (y \in B \ \vee \ y \in C) \\
&\iff (x \in A \wedge y \in B) \ \vee \ (x \in A \wedge y \in C) \qquad (\wedge \text{ dist. over } \vee) \\
&\iff (x, y) \in A \times B \ \vee \ (x, y) \in A \times C \\
&\iff (x, y) \in (A \times B) \cup (A \times C). \qquad \blacksquare
\end{aligned}$$

Identity 3: $A \times (B \cap C) = (A \times B) \cap (A \times C)$

Let (x, y) be an arbitrary ordered pair:

$$\begin{aligned}
(x,y) \in A \times (B \cap C) &\iff x \in A \ \wedge \ y \in B \cap C \\
&\iff x \in A \ \wedge \ y \in B \ \wedge \ y \in C \\
&\iff (x \in A \wedge y \in B) \ \wedge \ (x \in A \wedge y \in C) \quad \text{(idempotency of } \wedge) \\
&\iff (x,y) \in A \times B \ \wedge \ (x,y) \in A \times C \\
&\iff (x,y) \in (A \times B) \cap (A \times C). \qquad \blacksquare
\end{aligned}$$

Identity 5: $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

Let (x, y) be an arbitrary ordered pair:

$$\begin{aligned}
(x,y) \in A \times (B \setminus C) &\iff x \in A \ \wedge \ y \in B \ \wedge \ y \notin C \\
&\iff \underbrace{(x \in A \wedge y \in B)}_{(x,y) \in A \times B} \ \wedge \ \underbrace{\neg(x \in A \wedge y \in C)}_{(x,y) \notin A \times C} \qquad (\star) \\
&\iff (x,y) \in (A \times B) \setminus (A \times C).
\end{aligned}$$

Justification of $(\star)$*:* We need $(p \wedge q \wedge \neg r) \iff (p \wedge q) \wedge \neg(p \wedge r)$, where $p = (x \in A)$, $q = (y \in B)$, $r = (y \in C)$.

$(\Rightarrow)$ $p, q, \neg r$ all hold, so $p \wedge q$ holds. Since $\neg r$, the conjunction $p \wedge r$ is false, so $\neg(p \wedge r)$ holds.

$(\Leftarrow)$ From $p \wedge q$ we have p and q. From $\neg(p \wedge r)$ and p, we get $\neg r$. $\square$

Identities 2, 4, 6 are the mirror versions $(B \diamond C) \times A = (B \times A) \diamond (C \times A)$ for $\diamond \in \{\cup, \cap, \setminus\}$. Replace "$x \in A$" with "$y \in A$" throughout—the proofs are identical.

Practice 2–3 — Relations

Solutions to Selected Problems

Problems 1.2, 2.1, 2.2, 3.3, 4.1, 7.2

Discrete Mathematics — QYSU

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1 Problem 1.2 — Domain and Range of Relations

Theory Recap: Domain and Range

For a relation $\alpha \subseteq A \times B$:

- **Domain:** $D_\alpha = \{\, x \mid \exists\, y :\ (x, y) \in \alpha \,\}$ — the set of all first coordinates.
- **Range:** $R_\alpha = \{\, y \mid \exists\, x :\ (x, y) \in \alpha \,\}$ — the set of all second coordinates.

Problem Statement

Find the domain D_α and range R_α for each of the following relations.

(a) $\alpha = \{(a, 1),\ (a, 2),\ (c, 1),\ (c, 2),\ (c, 4),\ (d, 5)\}$

Reading off the first coordinates and second coordinates:

$$D_\alpha = \{a,\ c,\ d\}, \qquad R_\alpha = \{1,\ 2,\ 4,\ 5\}.$$

(b) $\alpha = \{(1, 2),\ (2, 4),\ (3, 6),\ (4, 8),\ \dots\}$

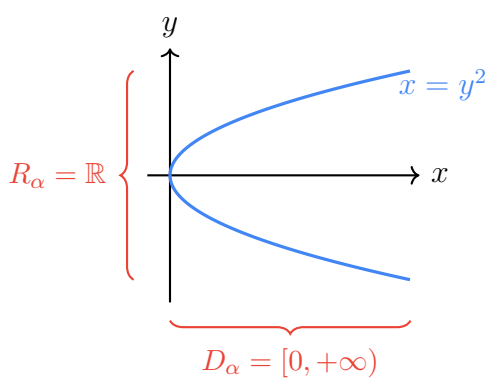
The pattern is $(n,\ 2n)$ for $n \in \mathbb{N}$, so:

$$D_\alpha = \mathbb{N} = \{1, 2, 3, \dots\}, \qquad R_\alpha = \{2, 4, 6, 8, \dots\} = 2\mathbb{N}.$$

(c) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x = y^2\}$

Since $x = y^2 \geq 0$ for all y, and for every $x \geq 0$ there exists $y = \pm\sqrt{x}$:

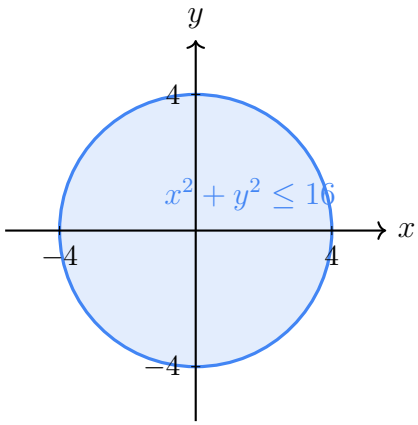
$$D_\alpha = [0, +\infty), \qquad R_\alpha = \mathbb{R}.$$



(d) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 16\}$

This is a closed disk of radius 4 centred at the origin. For any x with $|x| \leq 4$, there exists y satisfying $x^2 + y^2 \leq 16$, and vice versa.

$D_\alpha = [-4,\, 4], \qquad R_\alpha = [-4,\, 4].$



(e) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid x \mid y\}$

Here $x \mid y$ means "x divides y." Every $x \in \mathbb{N}$ divides some y (e.g. $y = x$), so $D_\alpha = \mathbb{N}$. Every $y \in \mathbb{N}$ is divided by some x (e.g. $x = 1$), so $R_\alpha = \mathbb{N}$.

$D_\alpha = \mathbb{N}, \quad R_\alpha = \mathbb{N}.$

(f) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid y \mid x\}$

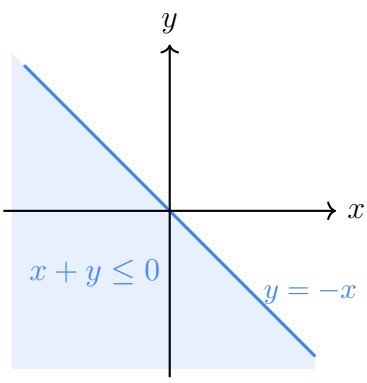
This is the inverse of part (e). By the same reasoning:

$D_\alpha = \mathbb{N}, \quad R_\alpha = \mathbb{N}.$

(g) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x + y \leq 0\}$

This is the half-plane below and including the line $y = -x$. For any $x \in \mathbb{R}$, choose $y = -x - 1$; then $x + y = -1 \leq 0$. For any $y \in \mathbb{R}$, choose $x = -y - 1$.

$D_\alpha = \mathbb{R}, \quad R_\alpha = \mathbb{R}.$



(h) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid 2x \geq 3y\}$

Equivalently $y \leq \frac{2}{3}x$. For any x, choose $y = 0$ (when $x \geq 0$) or any sufficiently small y. For any y, choose $x = \frac{3}{2}y$.

$D_\alpha = \mathbb{R}, \quad R_\alpha = \mathbb{R}.$

2 Problem 2.1 — Graph and Matrix of Relations

Theory Recap: Graphs and Matrices of Relations

Let α be a relation on a finite set $A = \{a_1, \ldots, a_n\}$.

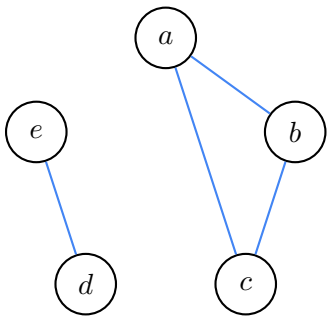
- **Graph:** Each element of A is a vertex. For a symmetric relation, if $(a_i, a_j) \in \alpha$ with $i \neq j$, draw an *undirected* edge between a_i and a_j.
- **Matrix:** The $n \times n$ matrix M with $M_{ij} = 1$ if $(a_i, a_j) \in \alpha$, and $M_{ij} = 0$ otherwise. For a symmetric relation $M = M^T$.

Problem Statement

Construct the graph and matrix for the following relations on $A = \{a, b, c, d, e\}$.

(a) $\alpha = \{(a,b), (b,a), (b,c), (c,b), (c,a), (a,c), (d,e), (e,d)\}$

Since every pair appears with its reverse, the relation is symmetric. Undirected edges: a–b, b–c, a–c, d–e.

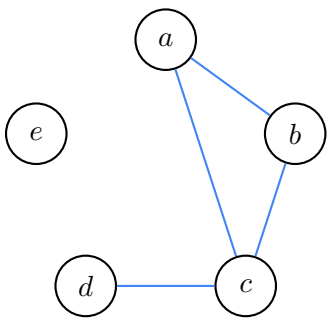


$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

Two connected components: the triangle $\{a, b, c\}$ and the edge $\{d, e\}$.

(b) $\alpha = \{(a,b), (b,a), (b,c), (c,b), (c,d), (d,c), (c,a), (a,c)\}$

Undirected edges: a–b, b–c, c–d, a–c.



$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

Triangle $\{a, b, c\}$ with d attached to c; vertex e is isolated.

(c)

Part (c) is identical to part (b) in the original problem sheet. The graph and matrix are the same as above.

3 Problem 2.2 — Digraph and Matrix of Relations

Theory Recap: Directed Graphs (Digraphs)

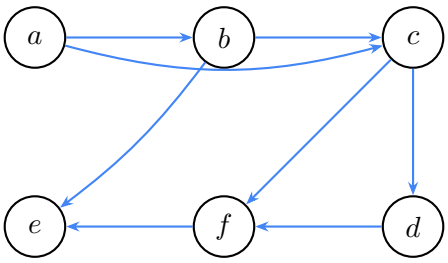
For a (not necessarily symmetric) relation α on A:

- **Digraph:** Vertices are elements of A. For each $(a_i, a_j) \in \alpha$, draw an arrow $a_i \to a_j$. A pair (a_i, a_i) is a *loop*.
- **Matrix:** Same construction; the matrix need not be symmetric.

Problem Statement

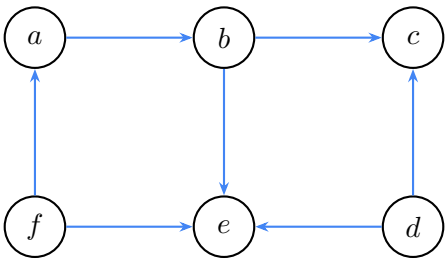
Construct the digraph and matrix for the following relations on $A = \{a, b, c, d, e, f\}$.

(a) $\alpha = \{(a,b), (b,c), (a,c), (b,e), (c,f), (c,d), (d,f), (f,e)\}$



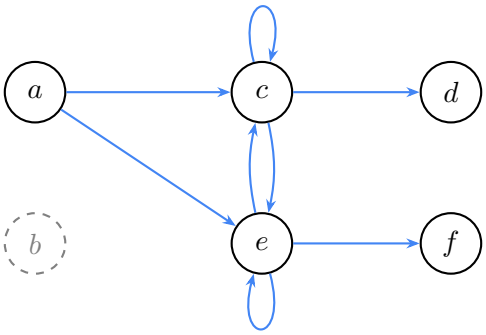
$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

(b) $\alpha = \{(a,b), (b,c), (d,c), (d,e), (f,e), (f,a), (b,e)\}$



$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array}$$

(c) $\alpha = \{(a,c),(c,c),(c,d),(c,e),(e,c),(e,e),(e,f),(a,e)\}$



Note: vertex b (dashed) is completely isolated — no edges touch it.

$$M_\alpha = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \end{array} \begin{array}{c} \begin{array}{cccccc} a & b & c & d & e & f \end{array} \\ \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

The loops at c and e appear as 1's on the main diagonal. The b-row and b-column are all zeros.

4 Problem 3.3 — Composition of Relations

Theory Recap: Composition of Relations

Let $\alpha \subseteq A \times B$ and $\beta \subseteq B \times C$. The **composition** $\alpha \circ \beta$ (also written $\alpha \cdot \beta$) is the relation on $A \times C$ defined by:

$$(x, z) \in \alpha \circ \beta \iff \exists\, y \in B : \ (x, y) \in \alpha \ \wedge \ (y, z) \in \beta.$$

Read left-to-right: first go through α (from x to some y), then through β (from y to z).

Problem Statement

Let α and β be relations on $\mathbb{Z}$:

$$\alpha = \{(x,\, x+2) \mid x \in \mathbb{Z}\}, \qquad \beta = \{(x,\, x^2) \mid x \in \mathbb{Z}\}.$$

Find $\alpha \circ \beta$ and $\beta \circ \alpha$.

Finding $\alpha \circ \beta$

We apply α first, then β:

- $(x, y) \in \alpha \implies y = x + 2$.
- $(y, z) \in \beta \implies z = y^2 = (x+2)^2$.

$$\alpha \circ \beta = \{\, (x,\ (x+2)^2) \mid x \in \mathbb{Z} \,\} = \{\, (x,\ x^2 + 4x + 4) \mid x \in \mathbb{Z} \,\}.$$

Finding $\beta \circ \alpha$

We apply β first, then α:

- $(x, y) \in \beta \implies y = x^2$.
- $(y, z) \in \alpha \implies z = y + 2 = x^2 + 2$.

$$\beta \circ \alpha = \{\, (x,\ x^2 + 2) \mid x \in \mathbb{Z} \,\}.$$

Verification and comparison

x	$\alpha \circ \beta$: $(x+2)^2$	$\beta \circ \alpha$: $x^2 + 2$
-2	0	6
0	4	2
1	9	3
3	25	11

Key observation: $\alpha \circ \beta \neq \beta \circ \alpha$, confirming that composition of relations is *not commutative* in general.

$$\begin{tikzcd} x \arrow[r, "\alpha"] & x+2 \arrow[r, "\beta"] & (x+2)^2 \end{tikzcd}$$

$$\alpha \circ \beta$$

$$\begin{tikzcd} x \arrow[r, "\beta"] & x^2 \arrow[r, "\alpha"] & x^2+2 \end{tikzcd}$$

$$\beta \circ \alpha$$

5 Problem 4.1 — Properties of Relations

Theory Recap: Properties of Relations

A relation α on a set A may satisfy:

- **Reflexive:** $\forall x \in A: \ (x, x) \in \alpha$.
- **Symmetric:** $(x, y) \in \alpha \ \Rightarrow \ (y, x) \in \alpha$.
- **Antisymmetric:** $(x, y) \in \alpha \ \wedge \ (y, x) \in \alpha \ \Rightarrow \ x = y$.
- **Transitive:** $(x, y) \in \alpha \ \wedge \ (y, z) \in \alpha \ \Rightarrow \ (x, z) \in \alpha$.

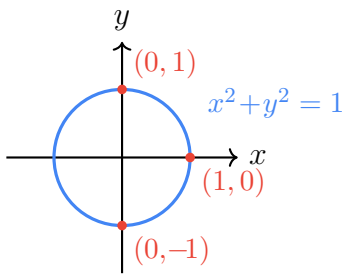
Note that symmetric and antisymmetric are *not* negations of each other — a relation can be both (e.g. Δ_A) or neither.

Problem Statement

Determine which properties each of the following relations possesses.

(a) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$

Geometrically, $(x, y) \in \alpha$ iff the point (x, y) lies on the unit circle.



Reflexive? No. We'd need $2x^2 = 1$ for *all* x, which fails for $x = 0$.

Symmetric? Yes. $x^2 + y^2 = y^2 + x^2$. ✓

Antisymmetric? No. $(0, 1) \in \alpha$ and $(1, 0) \in \alpha$, but $0 \neq 1$.

Transitive? No. $(1, 0)$ and $(0, 1) \in \alpha$, but $(1, 1) \notin \alpha$ since $1 + 1 = 2 \neq 1$.

Symmetric only.

(b) $\alpha = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y^2\}$

Equivalently $|x| = |y|$, so $y = \pm x$.

Reflexive? Yes. $x^2 = x^2$ always holds. ✓

Symmetric? Yes. $x^2 = y^2 \Rightarrow y^2 = x^2$. ✓

Antisymmetric? No. $1^2 = (-1)^2$ and $(-1)^2 = 1^2$, but $1 \neq -1$.

Transitive? Yes. $x^2 = y^2$ and $y^2 = z^2$ imply $x^2 = z^2$. ✓

Reflexive, symmetric, transitive — this is an **equivalence relation**.
Equivalence classes: $[x] = \{x, -x\}$ for each $x \in \mathbb{R}$.

(c) $\alpha = \{(x, y) \in \mathbb{Z}^2 \mid x \mid y\}$

Here $x \mid y$ means "x divides y," i.e. $\exists\, k \in \mathbb{Z} :\ y = kx$.

Reflexive? Yes. $x = 1 \cdot x$, so $x \mid x$ for all $x \in \mathbb{Z}$. ✓

Symmetric? No. $1 \mid 2$ but $2 \nmid 1$.

Antisymmetric? No (on $\mathbb{Z}$!). We have $2 \mid (-2)$ and $(-2) \mid 2$, but $2 \neq -2$.

Transitive? Yes. If $y = kx$ and $z = \ell y$, then $z = \ell k x$, so $x \mid z$. ✓

Reflexive and transitive, but *not* antisymmetric on $\mathbb{Z}$ (due to sign).

(d) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid x \mid y\}$

Same as (c) but restricted to $\mathbb{N} = \{1, 2, 3, \dots\}$ (positive integers).

Reflexive? Yes. $x = 1 \cdot x$. ✓

Symmetric? No. $1 \mid 2$ but $2 \nmid 1$.

Antisymmetric? Yes. If $x \mid y$ and $y \mid x$ with $x, y > 0$: then $y = kx$ and $x = \ell y$ with $k, \ell \geq 1$, so $x = k\ell\, x$ giving $k\ell = 1$, hence $k = \ell = 1$ and $x = y$. ✓

Transitive? Yes (same argument as (c)). ✓

Reflexive, antisymmetric, transitive — this is a **partial order**.
Compare with (c): restricting from $\mathbb{Z}$ to $\mathbb{N}$ restores antisymmetry.

(e) $\alpha = \{(x, y) \in \mathbb{N}^2 \mid \gcd(x, y) = 1\}$

$(x, y) \in \alpha$ iff x and y are coprime.

Reflexive? No. $\gcd(2, 2) = 2 \neq 1$.

Symmetric? Yes. $\gcd(x, y) = \gcd(y, x)$. ✓

Antisymmetric? No. $\gcd(2, 3) = 1$ and $\gcd(3, 2) = 1$, but $2 \neq 3$.

Transitive? No. $\gcd(2, 3) = 1$ and $\gcd(3, 4) = 1$, but $\gcd(2, 4) = 2 \neq 1$.

Symmetric only.

(f) $\alpha = \{(x, y) \in \mathbb{Z}^2 \mid \gcd(x, y) = 1\}$

The analysis is identical to part (e) since gcd depends only on absolute values. The same counterexamples apply.

Symmetric only.

Summary Table

Relation	Refl.	Symm.	Antisymm.	Trans.	Type
(a) $x^2 + y^2 = 1$ on $\mathbb{R}$	—	✓	—	—	—
(b) $x^2 = y^2$ on $\mathbb{R}$	✓	✓	—	✓	equivalence
(c) $x \mid y$ on $\mathbb{Z}$	✓	—	—	✓	preorder
(d) $x \mid y$ on $\mathbb{N}$	✓	—	✓	✓	partial order
(e) $\gcd = 1$ on $\mathbb{N}$	—	✓	—	—	—
(f) $\gcd = 1$ on $\mathbb{Z}$	—	✓	—	—	—

6 Problem 7.2 — Closures of a Relation

Theory Recap: Closures of a Relation

Let α be a relation on a finite set A.

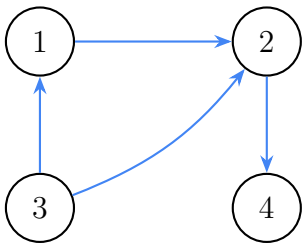
- **Reflexive closure** $r(\alpha) = \alpha \cup \Delta_A$, where $\Delta_A = \{(x, x) \mid x \in A\}$.
- **Symmetric closure** $s(\alpha) = \alpha \cup \alpha^{-1}$, where $\alpha^{-1} = \{(y, x) \mid (x, y) \in \alpha\}$.
- **Transitive closure** $t(\alpha) = \alpha \cup \alpha^2 \cup \alpha^3 \cup \cdots$ (union of all powers until no new pairs appear).

The closure adds the *fewest* pairs needed to acquire the property.

Problem Statement

Let $A = \{1, 2, 3, 4\}$ and $\alpha = \{(1, 2), (3, 1), (3, 2), (2, 4)\}$.
Find $r(\alpha)$, $s(\alpha)$, and $t(\alpha)$.

The original relation



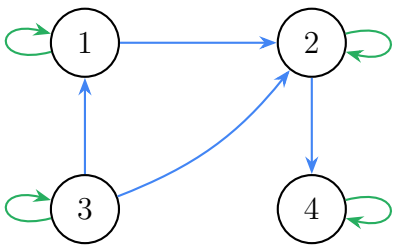
$$M_\alpha = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Rows/columns: $1, 2, 3, 4$.

Reflexive closure $r(\alpha)$

Add all diagonal pairs (x, x) for $x \in A$:

$$\begin{aligned} r(\alpha) &= \alpha \cup \{(1, 1), (2, 2), (3, 3), (4, 4)\} \\ &= \{(1, 1), (1, 2), (2, 2), (2, 4), (3, 1), (3, 2), (3, 3), (4, 4)\}. \end{aligned}$$



$$M_{r(\alpha)} = \begin{pmatrix} \mathbf{1} & 1 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 1 \\ 1 & 1 & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{pmatrix}$$

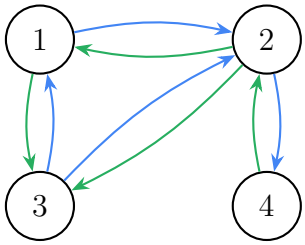
Bold = added 1's on diagonal.

Symmetric closure $s(\alpha)$

Add the reverse of every pair:

$$\alpha^{-1} = \{(2, 1), (1, 3), (2, 3), (4, 2)\}.$$

$$s(\alpha) = \alpha \cup \alpha^{-1} = \{(1, 2), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 2)\}.$$



$$M_{s(\alpha)} = \begin{pmatrix} 0 & 1 & \mathbf{1} & 0 \\ \mathbf{1} & 0 & \mathbf{1} & 1 \\ 1 & 1 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 \end{pmatrix}$$

Bold = added 1's (M is now symmetric).

Transitive closure $t(\alpha)$

We compute successive powers of α:

Step 1: $\alpha^2 = \alpha \circ \alpha$**.** For each chain $x \xrightarrow{\alpha} y \xrightarrow{\alpha} z$:

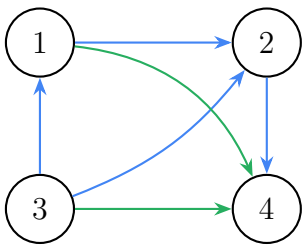
- $(1, 2)$ and $(2, 4)$: gives $\mathbf{(1, 4)}$. **New!**
- $(3, 1)$ and $(1, 2)$: gives $(3, 2)$. Already in α.
- $(3, 2)$ and $(2, 4)$: gives $\mathbf{(3, 4)}$. **New!**

New pairs: $(1, 4)$ and $(3, 4)$.

Step 2: $\alpha^3 = \alpha^2 \circ \alpha$**.** The new second coordinates are all 4, but 4 has no outgoing edges in α, so no further pairs arise.

$$t(\alpha) = \alpha \cup \alpha^2 = \{(1, 2),\ (1, 4),\ (2, 4),\ (3, 1),\ (3, 2),\ (3, 4)\}.$$

Interpretation: $(x, z) \in t(\alpha)$ iff there is a directed path from x to z in the original digraph.



$$M_{t(\alpha)} = \begin{pmatrix} 0 & 1 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Bold = added 1's.

Reachability summary:

- From 1: can reach 2 and 4 (path: $1 \to 2 \to 4$).
- From 2: can reach 4.
- From 3: can reach 1, 2, and 4 (path: $3 \to 1 \to 2 \to 4$).
- From 4: no outgoing edges — 4 is a *sink*.

Practice 4 — Functions (Mappings)

Solutions to Selected Problems

Problems 2, 4, 5, 6, 10, 12, 19, 22, 24, 25

Discrete Mathematics — QYSU

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1 Problem 2 — Domain and Range of Real Functions

Theory Recap: Domain and Range

For a function $f : \mathbb{R} \to \mathbb{R}$ given by a formula $f(x) = \ldots$:

- The **domain** $\mathrm{Dom}(f)$ is the set of all $x \in \mathbb{R}$ for which the formula is defined (no division by zero, no square root of a negative number, etc.).
- The **range** (image) $\mathrm{Ran}(f) = \{f(x) \mid x \in \mathrm{Dom}(f)\}$ is the set of all values actually attained by f.

Common restrictions:

Expression	Restriction	Reason
$\sqrt{g(x)}$	$g(x) \geq 0$	square root of negative undefined in $\mathbb{R}$
$1/g(x)$	$g(x) \neq 0$	division by zero undefined
$1/\sqrt{g(x)}$	$g(x) > 0$	both restrictions combined

Problem Statement

Let $f : \mathbb{R} \to \mathbb{R}$. Find the domain and range of:

(a) $f(x) = x^2 + 3$

(b) $f(x) = \sqrt{x+2}$

(c) $f(x) = \dfrac{1}{\sqrt{x-2}}$

(d) $f(x) = |x|$

(e) $f(x) = \dfrac{1}{x^2+2}$

(f) $f(x) = \dfrac{1}{x^2-2}$

Solution

(a) $f(x) = x^2 + 3$**.**

No restrictions on x: every real number can be squared.

$$\mathrm{Dom}(f) = \mathbb{R}.$$

Since $x^2 \geq 0$ for all $x \in \mathbb{R}$, we have $f(x) = x^2 + 3 \geq 3$. The minimum value 3 is attained at $x = 0$. As $x \to \pm\infty$, $f(x) \to +\infty$, and by continuity every value ≥ 3 is attained.

$$\mathrm{Ran}(f) = [3, +\infty).$$

$\mathrm{Dom}(f) = \mathbb{R}, \quad \mathrm{Ran}(f) = [3, +\infty).$

(b) $f(x) = \sqrt{x+2}$**.**

We need $x + 2 \geq 0$, i.e. $x \geq -2$.

$$\mathrm{Dom}(f) = [-2, +\infty).$$

At $x = -2$: $f(-2) = 0$. As $x \to +\infty$, $f(x) \to +\infty$. Since $\sqrt{\cdot}$ is continuous and non-negative:

$$\mathrm{Ran}(f) = [0, +\infty).$$

$$\boxed{\mathrm{Dom}(f) = [-2, +\infty), \quad \mathrm{Ran}(f) = [0, +\infty).}$$

(c) $f(x) = \dfrac{1}{\sqrt{x-2}}$.

We need $x - 2 > 0$ (strict inequality: the square root must be positive since it appears in the denominator), i.e. $x > 2$.

$$\mathrm{Dom}(f) = (2, +\infty).$$

As $x \to 2^+$: $\sqrt{x-2} \to 0^+$, so $f(x) \to +\infty$. As $x \to +\infty$: $\sqrt{x-2} \to +\infty$, so $f(x) \to 0^+$. Since f is continuous and strictly decreasing on $(2, +\infty)$, it takes every positive value:

$$\mathrm{Ran}(f) = (0, +\infty).$$

$$\boxed{\mathrm{Dom}(f) = (2, +\infty), \quad \mathrm{Ran}(f) = (0, +\infty).}$$

(d) $f(x) = |x|$.

The absolute value is defined for all real numbers.

$$\mathrm{Dom}(f) = \mathbb{R}.$$

Since $|x| \geq 0$ for all x, and $|0| = 0$, and $|x| \to +\infty$ as $x \to \pm\infty$:

$$\mathrm{Ran}(f) = [0, +\infty).$$

$$\boxed{\mathrm{Dom}(f) = \mathbb{R}, \quad \mathrm{Ran}(f) = [0, +\infty).}$$

(e) $f(x) = \dfrac{1}{x^2+2}$.

Since $x^2 + 2 \geq 2 > 0$ for all $x \in \mathbb{R}$, there is no division by zero.

$$\mathrm{Dom}(f) = \mathbb{R}.$$

Since $x^2 + 2 \geq 2$, we have $f(x) = \frac{1}{x^2+2} \leq \frac{1}{2}$. The maximum $\frac{1}{2}$ is attained at $x = 0$. As $x \to \pm\infty$, $f(x) \to 0^+$, but $f(x) > 0$ for all x. By continuity, every value in $(0, \frac{1}{2}]$ is attained.

$$\mathrm{Ran}(f) = \left(0, \tfrac{1}{2}\right].$$

$$\boxed{\mathrm{Dom}(f) = \mathbb{R}, \quad \mathrm{Ran}(f) = \left(0, \tfrac{1}{2}\right].}$$

(f) $f(x) = \dfrac{1}{x^2-2}$.

We need $x^2 - 2 \neq 0$, i.e. $x \neq \pm\sqrt{2}$.

$$\mathrm{Dom}(f) = \mathbb{R} \setminus \{-\sqrt{2},\ \sqrt{2}\}.$$

To find the range, set $y = \frac{1}{x^2-2}$ and solve for x^2:

$$y(x^2 - 2) = 1 \implies x^2 = \frac{1}{y} + 2 = \frac{1+2y}{y}.$$

We need $y \neq 0$ and $x^2 = \frac{1+2y}{y} \geq 0$.

Analyzing the sign of $\frac{1+2y}{y}$:

- $y > 0$: need $1 + 2y > 0$, which holds for all $y > 0$. So $y \in (0, +\infty)$.
- $y < 0$: need $1 + 2y < 0$ (numerator and denominator both negative), i.e. $y < -\frac{1}{2}$. So $y \in (-\infty, -\frac{1}{2})$.

At $x = 0$: $f(0) = \frac{1}{-2} = -\frac{1}{2}$, so $y = -\frac{1}{2}$ is attained.

$$\mathrm{Ran}(f) = \left(-\infty,\ -\tfrac{1}{2}\right] \cup (0, +\infty).$$

$$\boxed{\mathrm{Dom}(f) = \mathbb{R} \setminus \{-\sqrt{2},\ \sqrt{2}\}, \quad \mathrm{Ran}(f) = \left(-\infty,\ -\tfrac{1}{2}\right] \cup (0, +\infty).}$$

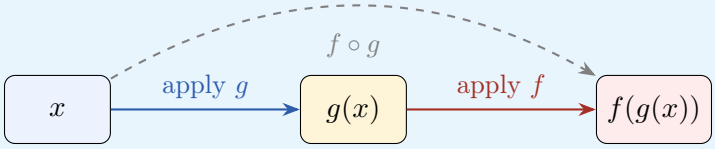
2 Problem 4 — Composition of Functions

Theory Recap: Function Composition

Given $f, g : \mathbb{R} \to \mathbb{R}$, the **composition** $f \circ g$ is defined by:

$$(f \circ g)(x) = f\big(g(x)\big).$$

Read right to left: first apply g, then apply f to the result.



Warning: In general, $f \circ g \neq g \circ f$ — composition is not commutative!

Problem Statement

Let $f, g : \mathbb{R} \to \mathbb{R}$. Find $f \circ g$ and $g \circ f$ if:

(a) $f(x) = x^2 + 1, \quad g(x) = x + 3$

(b) $f(x) = \sqrt{x^2 + 2}, \quad g(x) = x^2 + 3$

(c) $f(x) = \dfrac{1}{x}, \quad g(x) = 2x + 3$

Solution

(a) $f(x) = x^2 + 1$, $g(x) = x + 3$.

$$(f \circ g)(x) = f(g(x)) = f(x+3) = (x+3)^2 + 1 = x^2 + 6x + 9 + 1 = x^2 + 6x + 10.$$

$$(g \circ f)(x) = g(f(x)) = g(x^2+1) = (x^2+1) + 3 = x^2 + 4.$$

$$\boxed{(f \circ g)(x) = x^2 + 6x + 10, \qquad (g \circ f)(x) = x^2 + 4.}$$

(b) $f(x) = \sqrt{x^2 + 2}$, $g(x) = x^2 + 3$.

$$(f \circ g)(x) = f(g(x)) = f(x^2+3) = \sqrt{(x^2+3)^2 + 2} = \sqrt{x^4 + 6x^2 + 9 + 2} = \sqrt{x^4 + 6x^2 + 11}.$$

$$(g \circ f)(x) = g(f(x)) = g\left(\sqrt{x^2+2}\right) = \left(\sqrt{x^2+2}\right)^2 + 3 = x^2 + 2 + 3 = x^2 + 5.$$

$$\boxed{(f \circ g)(x) = \sqrt{x^4 + 6x^2 + 11}, \qquad (g \circ f)(x) = x^2 + 5.}$$

(c) $f(x) = \dfrac{1}{x}$, $g(x) = 2x + 3$.

$$(f \circ g)(x) = f(g(x)) = f(2x+3) = \frac{1}{2x+3}, \qquad x \neq -\tfrac{3}{2}.$$

$$(g \circ f)(x) = g(f(x)) = g\left(\frac{1}{x}\right) = \frac{2}{x} + 3 = \frac{2+3x}{x}, \qquad x \neq 0.$$

$$\boxed{(f \circ g)(x) = \frac{1}{2x+3}\ (x \neq -\tfrac{3}{2}), \qquad (g \circ f)(x) = \frac{2+3x}{x}\ (x \neq 0).}$$

3 Problem 5 — Inverse Functions

Theory Recap: Inverse Function

If $f : A \to B$ is a bijection, then f has an **inverse function** $f^{-1} : B \to A$ satisfying:

$$f^{-1}(y) = x \iff f(x) = y.$$

Algorithm to find f^{-1}:

1. Write $y = f(x)$.
2. Solve for x in terms of y.
3. Swap the names: the expression for x in terms of y is $f^{-1}(y)$.

Verification: $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Problem Statement

Find the inverse relation for:

(a) $f(x) = \dfrac{x+4}{2}$

(b) $f(x) = x^3$

(c) $f(x) = \dfrac{x-2}{x+3}$

Solution

(a) $f(x) = \dfrac{x+4}{2}$.

Set $y = \frac{x+4}{2}$. Solve for x:

$$2y = x + 4 \implies x = 2y - 4.$$

$$\boxed{f^{-1}(y) = 2y - 4.}$$

Verification: $f(f^{-1}(y)) = f(2y-4) = \frac{(2y-4)+4}{2} = \frac{2y}{2} = y$. ✓

$f^{-1}(f(x)) = f^{-1}\left(\frac{x+4}{2}\right) = 2 \cdot \frac{x+4}{2} - 4 = x + 4 - 4 = x$. ✓

$f^{-1}(x) = 2x - 4.$

(b) $f(x) = x^3$.

Set $y = x^3$. Solve for x:

$$x = \sqrt[3]{y} = y^{1/3}.$$

$$\boxed{f^{-1}(y) = \sqrt[3]{y}.}$$

Verification: $f(f^{-1}(y)) = (\sqrt[3]{y})^3 = y$. ✓ $\quad f^{-1}(f(x)) = \sqrt[3]{x^3} = x$. ✓

$f^{-1}(x) = \sqrt[3]{x}.$

(c) $f(x) = \dfrac{x-2}{x+3}, \qquad x \neq -3.$

Set $y = \frac{x-2}{x+3}$. Solve for x:

$$\begin{aligned} y(x+3) &= x-2 \\ yx + 3y &= x-2 \\ yx - x &= -2-3y \\ x(y-1) &= -(2+3y) \\ x &= \frac{-(2+3y)}{y-1} = \frac{2+3y}{1-y}, \qquad y \neq 1. \end{aligned}$$

Note that $f(x) = 1$ has no solution (it would require $x-2 = x+3$, i.e. $-2 = 3$), confirming $1 \notin \mathrm{Ran}(f)$.

$$\boxed{f^{-1}(y) = \frac{2+3y}{1-y}, \qquad y \neq 1.}$$

Verification: $f(f^{-1}(y)) = f\left(\frac{2+3y}{1-y}\right) = \frac{\frac{2+3y}{1-y}-2}{\frac{2+3y}{1-y}+3} = \frac{2+3y-2(1-y)}{2+3y+3(1-y)} = \frac{2+3y-2+2y}{2+3y+3-3y} = \frac{5y}{5} = y.$ ✓

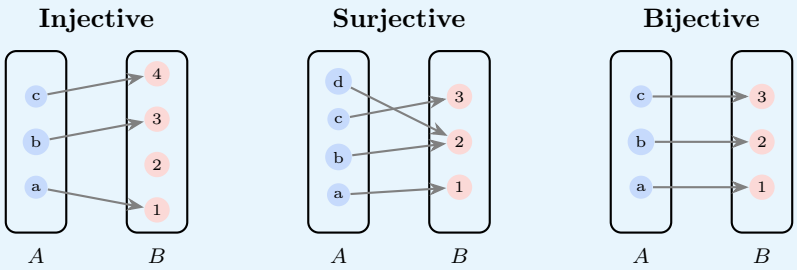
$$f^{-1}(x) = \frac{2+3x}{1-x}, \quad x \neq 1.$$

4 Problem 6 — Injectivity, Surjectivity, Bijectivity

Theory Recap: Injective, Surjective, Bijective

Let $f : A \to B$.

- **Injective** (one-to-one): $f(x_1) = f(x_2) \implies x_1 = x_2$. Equivalently, $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$. *Horizontal line test:* every horizontal line meets the graph at most once.
- **Surjective** (onto): $\forall\, y \in B,\ \exists\, x \in A : f(x) = y$. Equivalently, $\mathrm{Ran}(f) = B$.
- **Bijective**: both injective and surjective.



Problem Statement

Determine which of the following mappings $f : \mathbb{R} \to \mathbb{R}$ are injective, surjective, or bijective:

(a) $f(x) = |x|$
(b) $f(x) = x^2 + 4$
(c) $f(x) = x^3 + 6$
(d) $f(x) = x + |x|$
(e) $f(x) = x(x-2)(x+2)$

Solution

(a) $f(x) = |x|$.

Not injective: $f(1) = |1| = 1 = |-1| = f(-1)$, but $1 \neq -1$.

Not surjective: $|x| \geq 0$ for all x, so no x satisfies $f(x) = -1$. The range is $[0, +\infty) \neq \mathbb{R}$.

$f(x) = |x|$: neither injective nor surjective (hence not bijective).

(b) $f(x) = x^2 + 4$.

Not injective: $f(1) = 5 = f(-1)$, but $1 \neq -1$.

Not surjective: $x^2 + 4 \geq 4$ for all x, so no x satisfies $f(x) = 0$. The range is $[4, +\infty) \neq \mathbb{R}$.

$f(x) = x^2 + 4$: neither injective nor surjective (hence not bijective).

(c) $f(x) = x^3 + 6$.

Injective: Suppose $f(x_1) = f(x_2)$. Then $x_1^3 + 6 = x_2^3 + 6$, so $x_1^3 = x_2^3$. Since the cube function is strictly increasing on $\mathbb{R}$, this implies $x_1 = x_2$.

Surjective: Given any $y \in \mathbb{R}$, we need $x^3 + 6 = y$, i.e. $x = \sqrt[3]{y-6}$. Since the cube root is defined for all real numbers, such x always exists.

$f(x) = x^3 + 6$: **bijective** (both injective and surjective).

(d) $f(x) = x + |x|$**.**

First, let us simplify:

$$f(x) = x + |x| = \begin{cases} 2x & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

Not injective: $f(-1) = 0 = f(-2)$, but $-1 \neq -2$. (In fact, $f(x) = 0$ for all $x < 0$.)

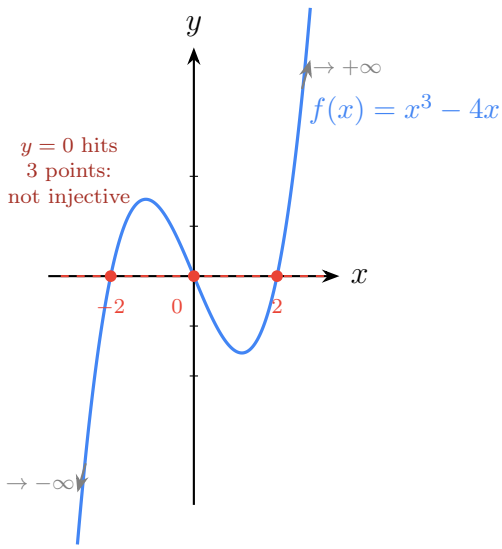
Not surjective: For $x \geq 0$, $f(x) = 2x \geq 0$. For $x < 0$, $f(x) = 0$. So $f(x) \geq 0$ for all x, and no x satisfies $f(x) = -1$. The range is $[0, +\infty) \neq \mathbb{R}$.

$f(x) = x + |x|$: neither injective nor surjective (hence not bijective).

(e) $f(x) = x(x-2)(x+2) = x^3 - 4x$**.**

Not injective: $f(0) = 0$ and $f(2) = 2 \cdot 0 \cdot 4 = 0$, so $f(0) = f(2) = 0$ but $0 \neq 2$.

Surjective: $f(x) = x^3 - 4x$ is a cubic polynomial with positive leading coefficient. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to +\infty$, $f(x) \to +\infty$. By the Intermediate Value Theorem, every $y \in \mathbb{R}$ is attained.



$f(x) = x(x-2)(x+2)$: surjective but **not injective** (hence not bijective).

5 Problem 10 — Kernel of a Mapping is an Equivalence Relation

Theory Recap: Kernel of a Mapping

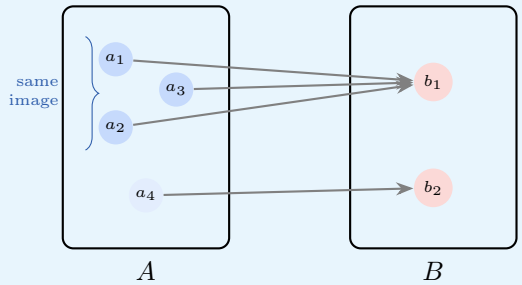
Let $f : A \to B$ be a mapping. The **kernel** of f is the relation on A defined by:

$$\mathrm{Ker}(f) = \{(a_1, a_2) \in A \times A \mid f(a_1) = f(a_2)\}.$$

We write $a_1 \sim a_2$ to mean $(a_1, a_2) \in \mathrm{Ker}(f)$, i.e. $f(a_1) = f(a_2)$.

An **equivalence relation** on A is a relation $\sim$ that is:

1. **Reflexive:** $a \sim a$ for all $a \in A$.
2. **Symmetric:** $a \sim b \implies b \sim a$.
3. **Transitive:** $a \sim b \wedge b \sim c \implies a \sim c$.



The equivalence classes of $\mathrm{Ker}(f)$ are exactly the **fibers** $f^{-1}(\{b\}) = \{a \in A \mid f(a) = b\}$ for $b \in \mathrm{Ran}(f)$.

Problem Statement

Prove that for any mapping $f : A \to B$, its kernel $\mathrm{Ker}(f)$ is an equivalence relation on A.

Solution

We must verify the three properties of an equivalence relation. Recall: $a_1 \sim a_2 \iff f(a_1) = f(a_2)$.

1. Reflexivity. Let $a \in A$. Then $f(a) = f(a)$ (equality is reflexive), so $a \sim a$. ✓

2. Symmetry. Suppose $a_1 \sim a_2$, i.e. $f(a_1) = f(a_2)$. Then $f(a_2) = f(a_1)$ (equality is symmetric), so $a_2 \sim a_1$. ✓

3. Transitivity. Suppose $a_1 \sim a_2$ and $a_2 \sim a_3$, i.e. $f(a_1) = f(a_2)$ and $f(a_2) = f(a_3)$. Then $f(a_1) = f(a_3)$ (equality is transitive), so $a_1 \sim a_3$. ✓

Since $\mathrm{Ker}(f)$ is reflexive, symmetric, and transitive, it is an equivalence relation on A. $\square$

The kernel $\mathrm{Ker}(f)$ inherits reflexivity, symmetry, and transitivity from the equality relation "=" on B, so it is an equivalence relation on A.

6 Problem 12 — $f(x) = \frac{x}{2x+1}$ on $\mathbb{Q}^+$: Injective but Not Surjective

Theory Recap: Injectivity and Surjectivity — Proof Techniques

To prove injectivity: Assume $f(x_1) = f(x_2)$ and show $x_1 = x_2$ (direct proof), or assume $x_1 \neq x_2$ and show $f(x_1) \neq f(x_2)$ (contrapositive).
To disprove surjectivity: Find a specific y in the codomain for which $f(x) = y$ has no solution in the domain.
To prove surjectivity: Given arbitrary y in the codomain, solve $f(x) = y$ and verify x lies in the domain.

Problem Statement

Let $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ be defined by $f(x) = \dfrac{x}{2x+1}$. Prove that f is injective but not surjective.

Solution

Injectivity. Let $x_1, x_2 \in \mathbb{Q}^+$ and suppose $f(x_1) = f(x_2)$:

$$\frac{x_1}{2x_1+1} = \frac{x_2}{2x_2+1}.$$

Cross-multiplying (the denominators are positive since $x_1, x_2 > 0$):

$$x_1(2x_2+1) = x_2(2x_1+1).$$

Expanding:

$$2x_1x_2 + x_1 = 2x_1x_2 + x_2.$$

Subtracting $2x_1x_2$ from both sides:

$$x_1 = x_2.$$

Therefore f is injective. $\square$

Not surjective. We claim that $y = \frac{1}{2}$ is in $\mathbb{Q}^+$ but not in $\mathrm{Ran}(f)$.
Suppose for contradiction that there exists $x \in \mathbb{Q}^+$ with $f(x) = \frac{1}{2}$:

$$\frac{x}{2x+1} = \frac{1}{2} \implies 2x = 2x+1 \implies 0 = 1.$$

This is a contradiction. Therefore no $x \in \mathbb{Q}^+$ satisfies $f(x) = \frac{1}{2}$.

More generally, for any $x > 0$:

$$f(x) = \frac{x}{2x+1} = \frac{1}{2+\frac{1}{x}} < \frac{1}{2},$$

since $\frac{1}{x} > 0$, so $2 + \frac{1}{x} > 2$, and thus $\frac{1}{2+\frac{1}{x}} < \frac{1}{2}$. Therefore $\mathrm{Ran}(f) \subseteq \left(0, \frac{1}{2}\right)$, so f is not surjective onto $\mathbb{Q}^+$. $\square$

$f(x) = \frac{x}{2x+1}$ on $\mathbb{Q}^+$ is **injective** (by cross-multiplication) but **not surjective** (e.g., $y = \frac{1}{2}$ is not in the range, since $f(x) < \frac{1}{2}$ for all $x > 0$).

7 Problem 19 — $f(A \cap B) = f(A) \cap f(B)$ Iff f Is Injective

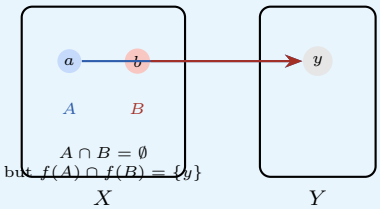
Theory Recap: Images of Intersections

For any mapping $f : X \to Y$ and subsets $A, B \subseteq X$:

$$f(A \cap B) \subseteq f(A) \cap f(B) \qquad \text{always holds.}$$

However, equality $f(A \cap B) = f(A) \cap f(B)$ does **not** hold in general. It holds for *all* $A, B \subseteq X$ if and only if f is **injective**.

Non-injective f



Problem Statement

For which mappings does $f(A \cap B) = f(A) \cap f(B)$ hold for all subsets A, B of the domain?
Answer: injective mappings. Prove both directions.

Solution

Direction 1: If f is injective, then $f(A \cap B) = f(A) \cap f(B)$.

The inclusion $f(A \cap B) \subseteq f(A) \cap f(B)$ holds for any function (if $x \in A \cap B$, then $f(x) \in f(A)$ and $f(x) \in f(B)$).

We prove the reverse inclusion $f(A) \cap f(B) \subseteq f(A \cap B)$.

Let $y \in f(A) \cap f(B)$. Then:

- $y \in f(A)$: there exists $a \in A$ with $f(a) = y$.
- $y \in f(B)$: there exists $b \in B$ with $f(b) = y$.

Since $f(a) = y = f(b)$ and f is injective, we conclude $a = b$. Call this common element $c = a = b$. Then $c \in A$ and $c \in B$, so $c \in A \cap B$. Therefore $y = f(c) \in f(A \cap B)$.

This shows $f(A) \cap f(B) \subseteq f(A \cap B)$, completing the proof of equality. $\square$

Direction 2: If $f(A \cap B) = f(A) \cap f(B)$ for all A, B, then f is injective.

We prove the contrapositive: if f is not injective, then there exist A, B with $f(A \cap B) \neq f(A) \cap f(B)$.

Suppose f is not injective. Then there exist $x_1 \neq x_2$ in the domain with $f(x_1) = f(x_2)$. Set $A = \{x_1\}$ and $B = \{x_2\}$. Then:

- $A \cap B = \emptyset$ (since $x_1 \neq x_2$), so $f(A \cap B) = f(\emptyset) = \emptyset$.

- $f(A) = \{f(x_1)\}$ and $f(B) = \{f(x_2)\} = \{f(x_1)\}$, so $f(A) \cap f(B) = \{f(x_1)\} \neq \emptyset$.

Therefore $f(A \cap B) \neq f(A) \cap f(B)$. $\square$

$f(A \cap B) = f(A) \cap f(B)$ holds for all A, B if and only if f is injective.

8 Problem 22 — Disjoint Subsets with Equal Images

Theory Recap: Images of Disjoint Sets

If $f : A \to B$ is **not** injective, then different elements can map to the same image. This means disjoint subsets of the domain can have overlapping — or even equal — images.

Problem Statement

Is it possible for $f : A \to B$ and disjoint non-empty $X_1, X_2 \subseteq A$ (i.e. $X_1 \cap X_2 = \emptyset$, $X_1 \neq \emptyset$, $X_2 \neq \emptyset$) to have $f(X_1) = f(X_2)$?

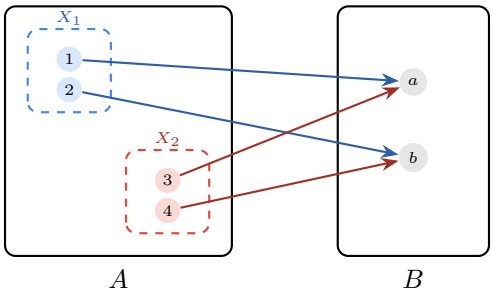
Solution

Yes, this is possible. We give a concrete example.

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b\}$, and define f by:

$$f(1) = a, \quad f(2) = b, \quad f(3) = a, \quad f(4) = b.$$

Take $X_1 = \{1, 2\}$ and $X_2 = \{3, 4\}$.



Then:

- $X_1 \cap X_2 = \{1, 2\} \cap \{3, 4\} = \emptyset$. ✓
- $f(X_1) = \{f(1), f(2)\} = \{a, b\}$.
- $f(X_2) = \{f(3), f(4)\} = \{a, b\}$.
- $f(X_1) = f(X_2) = \{a, b\}$. ✓

Yes. Disjoint non-empty subsets can have equal images when f is not injective. Example: $f(1) = a,\ f(2) = b,\ f(3) = a,\ f(4) = b$ with $X_1 = \{1, 2\}$, $X_2 = \{3, 4\}$.

9 Problem 24 — Preimages of Disjoint Sets

Theory Recap: Preimages Preserve Set Operations

For any function $f : A \to B$ and subsets $Y_1, Y_2 \subseteq B$, the **preimage** $f^{-1}(Y) = \{x \in A \mid f(x) \in Y\}$ satisfies:

- $f^{-1}(Y_1 \cup Y_2) = f^{-1}(Y_1) \cup f^{-1}(Y_2)$
- $f^{-1}(Y_1 \cap Y_2) = f^{-1}(Y_1) \cap f^{-1}(Y_2)$
- $f^{-1}(Y_1 \setminus Y_2) = f^{-1}(Y_1) \setminus f^{-1}(Y_2)$

These hold for *any* function — no injectivity or surjectivity is required!

Problem Statement

Is it true that for $f : A \to B$, if $Y_1 \cap Y_2 = \emptyset$, then $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \emptyset$?

Solution

Yes, this is always true.

Proof. Suppose $Y_1 \cap Y_2 = \emptyset$. We use the identity $f^{-1}(Y_1 \cap Y_2) = f^{-1}(Y_1) \cap f^{-1}(Y_2)$.

Since $Y_1 \cap Y_2 = \emptyset$:

$$f^{-1}(Y_1) \cap f^{-1}(Y_2) = f^{-1}(Y_1 \cap Y_2) = f^{-1}(\emptyset) = \emptyset.$$

Alternatively, we can prove it directly. Suppose for contradiction that there exists $x \in f^{-1}(Y_1) \cap f^{-1}(Y_2)$. Then:

- $x \in f^{-1}(Y_1) \implies f(x) \in Y_1$.
- $x \in f^{-1}(Y_2) \implies f(x) \in Y_2$.

So $f(x) \in Y_1 \cap Y_2 = \emptyset$, which is a contradiction since no element belongs to the empty set. $\square$

Yes. If $Y_1 \cap Y_2 = \emptyset$, then $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \emptyset$. This follows from the fact that preimages preserve intersections.

10 Problem 25 — $f(f^{-1}(Y)) \subseteq Y$

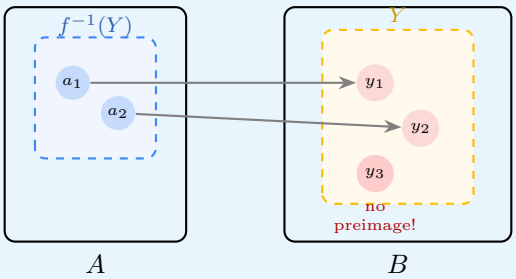
Theory Recap: Image of a Preimage

For $f : A \to B$ and $Y \subseteq B$:

$$f(f^{-1}(Y)) \subseteq Y \qquad \text{always holds.}$$

Equality $f(f^{-1}(Y)) = Y$ holds for all $Y \subseteq B$ if and only if f is **surjective**.

Intuition: $f^{-1}(Y)$ captures all elements in A that map into Y. Applying f again maps them back into Y — but some elements of Y may have *no preimage*, so they are "lost."



Here $f(f^{-1}(Y)) = \{y_1, y_2\} \subsetneq \{y_1, y_2, y_3\} = Y$.

Problem Statement

Prove that for $f : A \to B$ and $Y \subseteq B$:

$$f(f^{-1}(Y)) \subseteq Y.$$

Give an example where equality does not hold.

Solution

Proof of $f(f^{-1}(Y)) \subseteq Y$.

Let $y \in f(f^{-1}(Y))$. By definition of image, there exists $x \in f^{-1}(Y)$ such that $f(x) = y$. By definition of preimage, $x \in f^{-1}(Y)$ means $f(x) \in Y$. Therefore $y = f(x) \in Y$. $\square$

Example where equality fails.

Let $A = \{1\}$, $B = \{a, b\}$, and define $f : A \to B$ by $f(1) = a$. Take $Y = \{a, b\} = B$.

Then:

$$f^{-1}(Y) = f^{-1}(\{a, b\}) = \{x \in A \mid f(x) \in \{a, b\}\} = \{1\}.$$

$$f(f^{-1}(Y)) = f(\{1\}) = \{f(1)\} = \{a\}.$$

But $Y = \{a, b\}$. So:

$$f(f^{-1}(Y)) = \{a\} \subsetneq \{a, b\} = Y.$$

The element $b \in Y$ has no preimage under f (since f is not surjective), so it is "lost."

Proof: If $y \in f(f^{-1}(Y))$, then $y = f(x)$ for some $x \in f^{-1}(Y)$, which means $f(x) \in Y$, i.e. $y \in Y$.
Counterexample to equality: $A = \{1\}$, $B = \{a, b\}$, $f(1) = a$, $Y = B$. Then $f(f^{-1}(Y)) = \{a\} \subsetneq \{a, b\} = Y$.
Equality holds for all Y iff f is surjective.

Practice 5 — Combinatorics (Basic Rules)

Solutions to Selected Problems

§1: Problems 3, 5, 10 §2: Problems 5, 10, 12

Discrete Mathematics — QYSU

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Section 1: Fundamental Rules (Sum and Product Rules)

1 Problem 3 — Choosing Two Books of Different Subjects

Theory Recap: Sum and Product Rules

The two most fundamental counting principles:

- **Sum Rule (Rule of Alternatives):** If a task can be performed in one of k mutually exclusive ways, and way i can be done in n_i ways, then the total number of ways is $n_1 + n_2 + \cdots + n_k$.
- **Product Rule (Rule of Sequential Choices):** If a task consists of k sequential steps, where step i can be performed in n_i ways (independently of the choices in other steps), then the total number of ways is $n_1 \cdot n_2 \cdots n_k$.

Combined usage. When a task can be decomposed into several *cases* (sum rule), each of which involves *sequential choices* (product rule), we compute the count for each case using the product rule, then add the results.

Problem Statement

In how many ways can you choose two books of different subjects if there are 15 books on Informatics, 12 on Mathematics, and 10 on Physics?

Solution

Since the two chosen books must be on *different* subjects, we enumerate all possible pairs of subjects. There are three cases (sum rule), and within each case we apply the product rule:

Case	Subject 1	Subject 2	Count
Informatics + Mathematics	15 choices	12 choices	$15 \cdot 12 = 180$
Informatics + Physics	15 choices	10 choices	$15 \cdot 10 = 150$
Mathematics + Physics	12 choices	10 choices	$12 \cdot 10 = 120$

By the sum rule (the three cases are mutually exclusive), the total is:

$$15 \cdot 12 \;+\; 15 \cdot 10 \;+\; 12 \cdot 10 \;=\; 180 + 150 + 120 \;=\; 450.$$

There are $\boxed{450}$ ways to choose two books of different subjects.

2 Problem 5 — Natural Numbers Less Than 700 Divisible by 5 and/or 3

Theory Recap: Counting Multiples in a Range

The number of positive multiples of d that are less than or equal to N is $\left\lfloor \frac{N}{d} \right\rfloor$. More precisely, the multiples of d in $\{1, 2, \ldots, N\}$ are $d, 2d, 3d, \ldots, \lfloor N/d \rfloor \cdot d$, so there are exactly $\lfloor N/d \rfloor$ of them.

"Less than N" vs. "at most N". The natural numbers less than N are $\{1, 2, \ldots, N-1\}$. The number of multiples of d among these is $\left\lfloor \frac{N-1}{d} \right\rfloor$.

Problem Statement

How many natural numbers less than 700 are divisible by 5? How many of those are divisible by 3? How many are divisible by both 3 and 5?

Solution

We consider the set $S = \{1, 2, 3, \ldots, 699\}$ (natural numbers less than 700).

(a) Divisible by 5. The multiples of 5 in S are $5, 10, 15, \ldots, 695$. The number of such multiples is:

$$\left\lfloor \frac{699}{5} \right\rfloor = \lfloor 139.8 \rfloor = 139.$$

(b) Divisible by 3 (among natural numbers less than 700). The multiples of 3 in S are $3, 6, 9, \ldots, 699$. The number of such multiples is:

$$\left\lfloor \frac{699}{3} \right\rfloor = 233.$$

(c) Divisible by both 3 and 5. A number divisible by both 3 and 5 is divisible by $\mathrm{lcm}(3, 5) = 15$. The multiples of 15 in S are $15, 30, 45, \ldots, 690$. The number of such multiples is:

$$\left\lfloor \frac{699}{15} \right\rfloor = \lfloor 46.6 \rfloor = 46.$$

Divisible by 5: $\boxed{139}$. Divisible by 3: $\boxed{233}$. Divisible by both 3 and 5: $\boxed{46}$.

3 Problem 10 — 7-Digit Numbers with Alternating Parity

Theory Recap: Product Rule for Digit Counting

When counting numbers with digit-by-digit constraints, the product rule is applied position by position.

Key consideration: The *leading digit* (leftmost position) of an n-digit number **cannot be 0**, since that would reduce the number of digits. This restriction must always be accounted for separately.

Even digits: $\{0, 2, 4, 6, 8\}$ — five digits.
Odd digits: $\{1, 3, 5, 7, 9\}$ — five digits.

Problem Statement

Find the number of 7-digit numbers where the even-numbered positions contain odd digits and the odd-numbered positions contain even digits.

Solution

We label positions $1, 2, 3, 4, 5, 6, 7$ from left to right.

Pos. 1	Pos. 2	Pos. 3	Pos. 4	Pos. 5	Pos. 6	Pos. 7
odd (position)	*even* (position)	*odd* (position)	*even* (position)	*odd* (position)	*even* (position)	*odd* (position)
even digit	odd digit	even digit	odd digit	even digit	odd digit	even digit

Even positions (2, 4, 6): These must contain **odd** digits from $\{1, 3, 5, 7, 9\}$. Each position has 5 choices, giving 5^3 combinations.

Odd positions (1, 3, 5, 7): These must contain **even** digits from $\{0, 2, 4, 6, 8\}$. Each position has 5 choices *in general*, **except** position 1 (the leading digit), which **cannot be 0**.

- Position 1: $\{2, 4, 6, 8\}$ — **4 choices**.
- Positions 3, 5, 7: $\{0, 2, 4, 6, 8\}$ — 5 choices each, giving 5^3.

Applying the product rule. Since the choices at each position are independent:

$$\underbrace{4}_{\text{pos. 1}} \times \underbrace{5^3}_{\text{pos. 2, 4, 6}} \times \underbrace{5^3}_{\text{pos. 3, 5, 7}} = 4 \times 125 \times 125 = 4 \times 15{,}625 = 62{,}500.$$

There are $\boxed{62{,}500}$ such 7-digit numbers.

Section 2: Permutations and Arrangements

4 Problem 5 — Boys and Girls in a Line

Theory Recap: Permutations with Positional Constraints

When certain positions in a line are constrained (e.g., "the first and last positions must be occupied by specific types of people"), we use the following strategy:

1. **Fill the constrained positions first**, counting the number of ways to choose who goes there.
2. **Fill the remaining positions** with the remaining people, who can stand in any order.
3. **Multiply** the counts (product rule).

Recall that the number of ways to arrange n distinct objects in a line is $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$.

Problem Statement

In how many ways can 6 boys and 7 girls stand in a line such that the first and last positions are occupied by boys?

Solution

There are $6 + 7 = 13$ people in total.

Step 1: Fill the constrained positions.

- **First position:** must be a boy. There are 6 boys to choose from, so 6 choices.
- **Last position:** must be a boy (different from the first). There are $6 - 1 = 5$ remaining boys, so 5 choices.

Step 2: Fill the remaining positions. After placing two boys at the ends, there are $13 - 2 = 11$ people left (4 boys and 7 girls). These 11 people can stand in the remaining 11 positions (positions 2 through 12) in any order:

$$11! \text{ ways.}$$

Step 3: Apply the product rule. The choices at each step are independent, so:

$$6 \times 5 \times 11! \;=\; 30 \times 11!.$$

The total number of arrangements is $\boxed{30 \times 11! = 30 \times 39{,}916{,}800 = 1{,}197{,}504{,}000}$.

5 Problem 10 — Ramsey's Theorem $R(3,3) = 6$

Theory Recap: Pigeonhole Principle and Ramsey Theory

Pigeonhole Principle. If n objects are placed into k boxes and $n > k$, then at least one box contains more than one object.
More generally: if n objects are placed into k boxes, then at least one box contains $\lceil n/k \rceil$ objects.

Ramsey Theory (informal statement). In any sufficiently large structure, order must appear. The Ramsey number $R(s,t)$ is the smallest n such that every 2-colouring (say, red and blue) of the edges of the complete graph K_n contains either a red K_s or a blue K_t.

The classical result $R(3,3) = 6$ states: in any group of 6 people, there are always 3 mutual friends or 3 mutual strangers.

Problem Statement

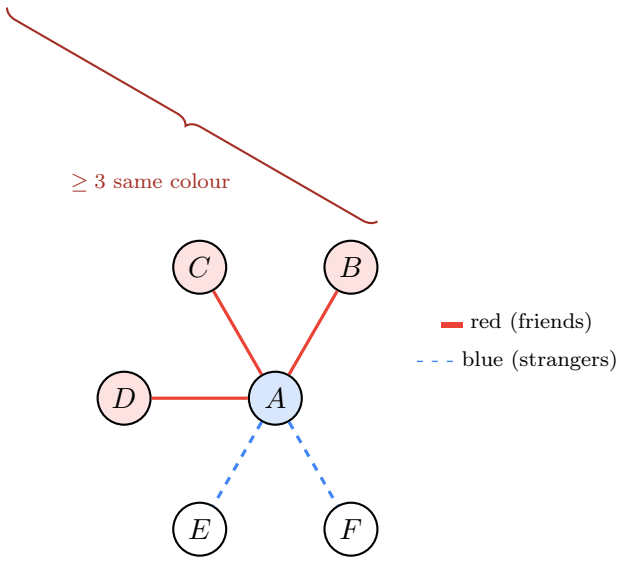
Prove that in any group of 6 people, there are either 3 mutual friends or 3 mutual strangers.

Solution (Proof)

We model the situation as a 2-colouring of the complete graph K_6. Each person is a vertex, and each edge is coloured **red** (friends) or **blue** (strangers). We must show that there exists a monochromatic triangle.

Step 1: Apply the Pigeonhole Principle. Pick any vertex A. Vertex A is connected to the other 5 vertices by 5 edges, each coloured red or blue. By the pigeonhole principle (5 edges into 2 colours), at least $\lceil 5/2 \rceil = 3$ edges from A have the same colour.

Without loss of generality, assume A has at least 3 **red** edges. Let the other endpoints of these red edges be B, C, D. (The case with 3 blue edges is handled symmetrically by swapping "friend" and "stranger.")



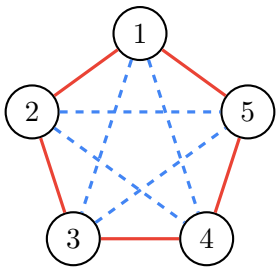
Step 2: Examine the edges among B, C, D. Consider the three edges BC, BD, CD. There are two cases:

Case 1: At least one of BC, BD, CD is **red**.
Suppose (WLOG) BC is red. Then A, B, C form a red triangle: edges AB, AC, and BC are all red. Thus we have 3 mutual **friends**.

Case 2: None of BC, BD, CD is red, i.e., all three are **blue**.
Then B, C, D form a blue triangle: edges BC, BD, CD are all blue. Thus we have 3 mutual **strangers**.

In both cases, a monochromatic triangle exists. □

Why 5 people are not enough. To show that $R(3,3) > 5$, we exhibit a 2-colouring of K_5 with no monochromatic triangle: arrange 5 vertices in a regular pentagon and colour the edges of the pentagon red and the diagonals blue (or vice versa). One can verify that no triangle is monochromatic.



K_5: edges red, diagonals blue — no monochromatic triangle

In any group of 6 people, there are always 3 mutual friends or 3 mutual strangers.
This proves $R(3,3) = 6$.

6 Problem 12 — Divisibility Among $7, 77, 777, \ldots$

Theory Recap: Pigeonhole Principle for Divisibility

A powerful application of the pigeonhole principle in number theory:

If we have $n+1$ integers and consider their remainders modulo n, then by the pigeonhole principle, at least two of them share the same remainder. Their *difference* is therefore divisible by n.

Repunit-like sequences. The sequence $7, 77, 777, 7777, \ldots$ can be written in closed form. The k-th term is:

$$a_k = \underbrace{77\ldots7}_{k \text{ sevens}} = 7 \cdot \underbrace{11\ldots1}_{k} = 7 \cdot \frac{10^k - 1}{9}.$$

Problem Statement

Prove that among the first 2014 terms of the sequence 7, 77, 777, 7777, ... there is one divisible by 2013.

Solution (Proof)

Let $a_k = \underbrace{77\ldots7}_{k}$ for $k = 1, 2, \ldots, 2014$.

Consider the 2014 remainders $r_k = a_k \bmod 2013$ for $k = 1, 2, \ldots, 2014$.

Case 1: Some $r_k = 0$. If $r_k = 0$ for some k, then a_k is divisible by 2013, and we are done.

Case 2: No $r_k = 0$. Then each $r_k \in \{1, 2, \ldots, 2012\}$. We have 2014 remainders taking values in a set of size 2012. By the **pigeonhole principle**, there exist indices $i < j$ (with $1 \leq i < j \leq 2014$) such that $r_i = r_j$, i.e.,

$$a_j \equiv a_i \pmod{2013}.$$

Therefore $2013 \mid (a_j - a_i)$.

Analysing $a_j - a_i$. We compute:

$$a_j - a_i = \underbrace{77\ldots7}_{j} - \underbrace{77\ldots7}_{i} = \underbrace{77\ldots7}_{j-i} \cdot 10^i = a_{j-i} \cdot 10^i.$$

So $2013 \mid a_{j-i} \cdot 10^i$.

Now, $2013 = 3 \times 11 \times 61$. Since $\gcd(10, 2013) = 1$ (as 2013 has no factors of 2 or 5), it follows that $\gcd(10^i, 2013) = 1$.

Therefore $2013 \mid a_{j-i}$.

Since $1 \leq j - i \leq 2013 < 2014$, the term a_{j-i} is indeed among the first 2014 terms of the sequence.

In both cases, there exists a term in $\{a_1, a_2, \ldots, a_{2014}\}$ that is divisible by 2013. $\square$

By the pigeonhole principle, among $a_1, a_2, \ldots, a_{2014}$, there is at least one term divisible by 2013.

Practice 6 — Combinations and Binomial Coefficients

Solutions to Selected Problems

Problems 8, 9, 11

Discrete Mathematics — QYSU

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1 Problem 8 — Lattice Paths Through a Given Point **2**

2 Problem 9 — Partitioning 6 into Three Summands **4**

3 Problem 11 — Rearrangements of "matematika" **6**

1 Problem 8 — Lattice Paths Through a Given Point

Theory Recap: Lattice Paths and Combinations

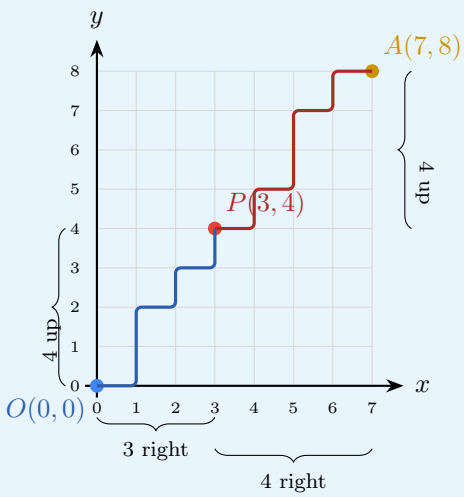
A **lattice path** from $(0,0)$ to (m,n) consists of exactly m steps right (R) and n steps up (U), in some order. The total number of steps is $m+n$, and we must choose which m of them are R-steps (the rest are U-steps):

$$\text{Number of paths from } (0,0) \text{ to } (m,n) = \binom{m+n}{m} = \binom{m+n}{n}.$$

Paths through an intermediate point. If a path must pass through a point $P(p,q)$ on the way from $O(0,0)$ to $A(m,n)$ (with $0 \le p \le m$ and $0 \le q \le n$), we decompose the journey into two independent segments:

$$O \to P: \binom{p+q}{p} \text{ paths}, \qquad P \to A: \binom{(m-p)+(n-q)}{m-p} \text{ paths}.$$

By the **multiplication principle**, the total count is the product.



Problem Statement

Find the number of lattice paths from $O(0,0)$ to $A(7,8)$ that pass through the point $P(3,4)$.

Solution

We split the path into two independent segments at $P(3,4)$.

Segment 1: $O(0,0) \to P(3,4)$. We need 3 steps right and 4 steps up, for a total of 7 steps. Choose which 3 are right-steps:

$$\binom{3+4}{3} = \binom{7}{3} = \frac{7!}{3! \cdot 4!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

Segment 2: $P(3,4) \to A(7,8)$. From $(3,4)$ to $(7,8)$ we need $7-3=4$ steps right and

$8 - 4 = 4$ steps up, for a total of 8 steps. Choose which 4 are right-steps:

$$\binom{4+4}{4} = \binom{8}{4} = \frac{8!}{4! \cdot 4!} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} = 70.$$

Multiplication principle. The two segments are independent (any path for the first segment can be combined with any path for the second), so:

$$\text{Total paths} = 35 \times 70 = 2450.$$

The number of lattice paths from $O(0,0)$ to $A(7,8)$ passing through $(3,4)$ is $\binom{7}{3} \cdot \binom{8}{4} = 35 \cdot 70 = \boxed{2450}$.

2 Problem 9 — Partitioning 6 into Three Summands

Theory Recap: Stars and Bars / Integer Compositions and Partitions

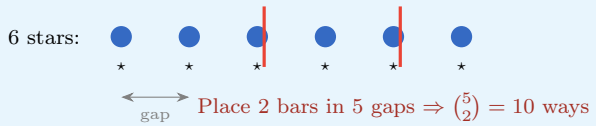
Stars and bars (combinations with repetition). The number of solutions to

$$x_1 + x_2 + \cdots + x_k = n, \qquad x_i \geq 0,$$

in non-negative integers is $\binom{n+k-1}{k-1}$.

If we require $x_i \geq 1$ (positive integers), substitute $y_i = x_i - 1 \geq 0$ to get $y_1 + \cdots + y_k = n - k$, giving $\binom{n-1}{k-1}$ solutions.

Example: $x_1 + x_2 + x_3 = 6$, $x_i \geq 1$



Integer partitions (unordered). A partition of n into exactly k parts is a way to write $n = \lambda_1 + \lambda_2 + \cdots + \lambda_k$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 1$. Order does not matter, so $4 + 1 + 1$ and $1 + 4 + 1$ are the *same* partition.

Problem Statement

Find the number of ways to partition the number 6 into three summands.

Solution

This problem admits two natural interpretations depending on whether the order of summands matters.

Interpretation 1: Ordered summands (compositions)

We seek the number of solutions to

$$x_1 + x_2 + x_3 = 6, \qquad x_i \geq 1.$$

By the stars-and-bars formula with the substitution $y_i = x_i - 1$:

$$\binom{6-1}{3-1} = \binom{5}{2} = \frac{5!}{2! \cdot 3!} = 10.$$

#	x_1	x_2	x_3
1	4	1	1
2	1	4	1
3	1	1	4
4	3	2	1
5	3	1	2
6	2	3	1
7	1	3	2
8	2	1	3
9	1	2	3
10	2	2	2

Interpretation 2: Unordered summands (integer partitions)

We seek the number of ways to write $6 = \lambda_1 + \lambda_2 + \lambda_3$ with $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 1$.

Enumerating all possibilities:

#	**Partition**
1	$4+1+1$
2	$3+2+1$
3	$2+2+2$

There are exactly **3** such partitions. We verify completeness: $\lambda_1 \geq 2$ always (since $\lambda_1 \geq \lceil 6/3 \rceil = 2$). If $\lambda_1 = 4$: $\lambda_2 + \lambda_3 = 2$, so $\lambda_2 = \lambda_3 = 1$. If $\lambda_1 = 3$: $\lambda_2 + \lambda_3 = 3$ with $3 \geq \lambda_2 \geq \lambda_3 \geq 1$, giving $\lambda_2 = 2, \lambda_3 = 1$. If $\lambda_1 = 2$: $\lambda_2 + \lambda_3 = 4$ with $2 \geq \lambda_2 \geq \lambda_3 \geq 1$, giving $\lambda_2 = 2, \lambda_3 = 2$.

Ordered (compositions with positive parts): $\binom{5}{2} = 10$ ways.

Unordered (integer partitions into 3 parts): 3 ways: 4+1+1, 3+2+1, 2+2+2.

3 Problem 11 — Rearrangements of "matematika"

Theory Recap: Multinomial Coefficient / Permutations with Repetition

If we have n objects where there are k distinct types with multiplicities $n_1, n_2, \ldots, n_k$ (so $n_1 + n_2 + \cdots + n_k = n$), the number of distinct arrangements is the **multinomial coefficient**:

$$\binom{n}{n_1,\ n_2,\ \ldots,\ n_k} = \frac{n!}{n_1! \cdot n_2! \cdots n_k!}.$$

Intuition. Start with $n!$ arrangements of all objects treated as distinguishable. Then divide out the $n_i!$ internal permutations for each group of identical objects.

$$\boxed{n!\text{ total arrangements}} \quad \div \quad \boxed{n_1!} \quad \boxed{n_2!} \quad \cdots \quad \boxed{n_k!} \quad = \quad \boxed{\text{distinct arrangements}}$$

Problem Statement

How many different words can be formed by rearranging all the letters of the word **"matematika"**?

Solution

Step 1: Identify the letters and their frequencies.

The word "matematika" has the letters:

$$\text{m} - \text{a} - \text{t} - \text{e} - \text{m} - \text{a} - \text{t} - \text{i} - \text{k} - \text{a}$$

This gives $n = 10$ letters in total. Counting each:

Letters:	**m**	**a**	**t**	**e**	**m**	**a**	**t**	**i**	**k**	**a**
Position:	1	2	3	4	5	6	7	8	9	10

Letter	a	m	t	e	i	k
Count n_i	3	2	2	1	1	1

Verification: $3 + 2 + 2 + 1 + 1 + 1 = 10$ ✓

Step 2: Apply the multinomial coefficient.

$$\frac{10!}{3! \cdot 2! \cdot 2! \cdot 1! \cdot 1! \cdot 1!} = \frac{10!}{3! \cdot 2! \cdot 2!}$$

Computing:

$$\begin{aligned} 10! &= 3\,628\,800, \\ 3! \cdot 2! \cdot 2! &= 6 \cdot 2 \cdot 2 = 24, \\ \frac{3\,628\,800}{24} &= 151\,200. \end{aligned}$$

The number of distinct rearrangements of "matematika" is $\dfrac{10!}{3! \cdot 2! \cdot 2!} = \boxed{151\,200}$.

Practice 7 — Partitions & Inclusion-Exclusion

Solutions to Selected Problems

Problems 1, 7, 10, 12

Discrete Mathematics — QYSU

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1 Problem 1 — Partitions of a 5-Element Set into 3 Classes

Theory Recap: Stirling Numbers of the Second Kind

The **Stirling number of the second kind** $S(n,k)$ counts the number of ways to partition an n-element set into exactly k non-empty, pairwise disjoint subsets (called *classes* or *blocks*).

Recurrence relation:

$$S(n,k) = S(n-1,\, k-1) \;+\; k \cdot S(n-1,\, k), \qquad n \geq 1,\; 1 \leq k \leq n.$$

Interpretation: Consider element n.

- *n is alone in its own class:* the remaining $n-1$ elements form $k-1$ classes $\longrightarrow$ $S(n-1,\, k-1)$ ways.
- *n joins one of the existing k classes:* first partition the other $n-1$ elements into k classes ($S(n-1,\, k)$ ways), then choose which class n joins (k choices) $\longrightarrow$ $k \cdot S(n-1,\, k)$ ways.

Boundary conditions: $S(n,1) = 1,\;\; S(n,n) = 1,\;\; S(n,0) = 0$ for $n \geq 1$.

$n \backslash k$	1	2	3	4	5
1	1				
2	1	1			
3	1	3	1		
4	1	7	6	1	
5	1	15	25 $\longleftarrow S(5,3) = 25$	10	1

Problem Statement

In how many ways can a 5-element set be partitioned into 3 non-empty classes?

Solution

We need the Stirling number $S(5,3)$. Using the recurrence:

$$S(5,3) = S(4,2) + 3 \cdot S(4,3).$$

Computing $S(4,2)$:

$$S(4,2) = S(3,1) + 2 \cdot S(3,2) = 1 + 2 \cdot 3 = 7.$$

Computing $S(4,3)$:

$$S(4,3) = S(3,2) + 3 \cdot S(3,3) = 3 + 3 \cdot 1 = 6.$$

Final result:

$$S(5,3) = 7 + 3 \cdot 6 = 7 + 18 = 25.$$

Verification by explicit enumeration. Let $\{a, b, c, d, e\}$ be our set. A partition into 3 classes must have type $\{3, 1, 1\}$ or $\{2, 2, 1\}$ (the only integer partitions of 5 into 3 parts).

- **Type $\{3, 1, 1\}$:** Choose the 3-element block: $\binom{5}{3} = 10$ ways. The remaining 2 singletons form an unordered pair, giving 1 arrangement. Total: 10.
- **Type $\{2, 2, 1\}$:** Choose the singleton: $\binom{5}{1} = 5$ ways. Partition the remaining 4 elements into two pairs: $\frac{1}{2!}\binom{4}{2} = 3$ ways (divide by 2! since the two pairs are unordered). Total: $5 \times 3 = 15$.

Grand total: $10 + 15 = 25$. ✓

$$S(5,3) = \boxed{25}$$

2 Problem 7 — Divisibility with Inclusion-Exclusion

Theory Recap: Inclusion-Exclusion Principle

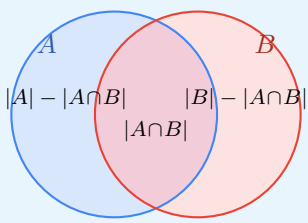
For events (or sets) $A_1, A_2, \ldots, A_m$ in a finite universe:

$$\left|\bigcup_{i=1}^{m} A_i\right| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots$$

To count elements that belong to **none** of $A_1, \ldots, A_m$:

$$\left|\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_m}\right| = |U| - \left|\bigcup_{i=1}^{m} A_i\right|.$$

Divisibility application: The number of integers in $\{1, 2, \ldots, N\}$ divisible by d is $\lfloor N/d \rfloor$. If $\gcd(d_1, d_2) = 1$, then divisibility by both d_1 and d_2 is the same as divisibility by $d_1 d_2$.



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Problem Statement

How many integers between 1 and 16500 are not divisible by 5 and not divisible by 3, but **are** divisible by 11?

Solution

Let S be the set of integers in $\{1, 2, \ldots, 16500\}$ that are divisible by 11. We want to remove from S those elements also divisible by 3 or by 5.

Define:

$$\begin{aligned} A &= \{n \in S \mid 3 \mid n\} && \text{(divisible by both 11 and 3, i.e. by 33)}, \\ B &= \{n \in S \mid 5 \mid n\} && \text{(divisible by both 11 and 5, i.e. by 55)}. \end{aligned}$$

We seek $|S| - |A \cup B|$.

Step 1: Compute individual counts.

$$\begin{aligned} |S| &= \lfloor 16500/11 \rfloor &&= 1500, \\ |A| &= \lfloor 16500/33 \rfloor &&= 500, \\ |B| &= \lfloor 16500/55 \rfloor &&= 300, \\ |A \cap B| &= \lfloor 16500/165 \rfloor &&= 100. \end{aligned}$$

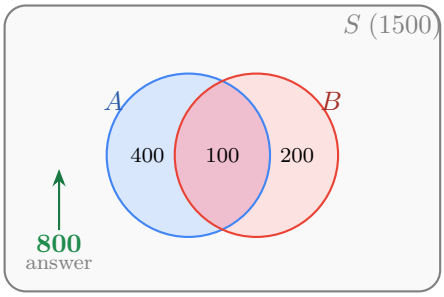
Here $|A \cap B|$ counts integers divisible by $\text{lcm}(33, 55) = 165$.

Step 2: Apply inclusion-exclusion.

$$|A \cup B| = |A| + |B| - |A \cap B| = 500 + 300 - 100 = 700.$$

Step 3: Subtract.

$$|S \setminus (A \cup B)| = |S| - |A \cup B| = 1500 - 700 = 800.$$



$\boxed{800}$ integers between 1 and 16500 are divisible by 11 but not by 3 and not by 5.

3 Problem 10 — Triangles in a Triangular Grid

Theory Recap: Counting Triangles in a Triangular Grid

When each side of an equilateral triangle is divided into n equal segments, connecting the division points with lines parallel to all three sides creates a **triangular grid** of side n.

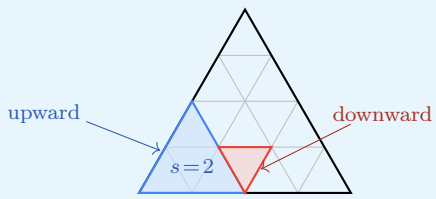
The grid contains two types of triangles with sides parallel to the original:

1. **Upward-pointing** triangles (same orientation as the original) of side length s ($1 \le s \le n$):

$$U(n,s) = \frac{(n-s+1)(n-s+2)}{2} = \binom{n-s+2}{2}.$$

2. **Downward-pointing** triangles (inverted orientation) of side length s ($1 \le s \le \lfloor n/2 \rfloor$):

$$D(n,s) = \frac{(n-2s+1)(n-2s+2)}{2} = \binom{n-2s+2}{2}.$$



Grid with $n = 4$

Problem Statement

Each side of triangle ABC is divided into 8 equal segments. Lines through the division points parallel to the sides create a triangular grid. How many triangles can be formed with vertices at these grid points such that their sides are parallel to the sides of ABC?

Solution

We have a triangular grid with $n = 8$. We count upward-pointing and downward-pointing triangles separately.

Upward-pointing triangles of side s ($1 \le s \le 8$):

$$U(8,s) = \frac{(8-s+1)(8-s+2)}{2} = \binom{10-s}{2}.$$

s	1	2	3	4	5	6	7	8	Total
$U(8,s)$	36	28	21	15	10	6	3	1	**120**

$$\sum_{s=1}^{8} U(8,s) = 36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 120.$$

Downward-pointing triangles of side s $(1 \le s \le \lfloor 8/2 \rfloor = 4)$:

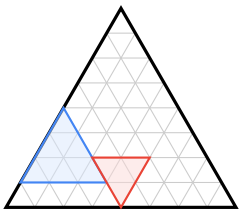
$$D(8,s) = \frac{(8-2s+1)(8-2s+2)}{2} = \binom{10-2s}{2}.$$

s	1	2	3	4	Total
$D(8,s)$	28	15	6	1	**50**

$$\sum_{s=1}^{4} D(8,s) = 28 + 15 + 6 + 1 = 50.$$

Grand total:

$$120 + 50 = 170.$$



$n = 8$ **grid**

The total number of triangles with sides parallel to ABC is $\boxed{170}$ (120 upward + 50 downward).

4 Problem 12 — Bounded Integer Solutions

Theory Recap: Counting Integer Solutions with Upper Bounds

To count non-negative integer solutions to $y_1 + y_2 + \cdots + y_r = n$ with upper bounds $y_i \le u_i$:

1. **Without upper bounds:** The number of solutions is $\binom{n+r-1}{r-1}$ (stars and bars).
2. **Subtract violations** using inclusion-exclusion. Let A_i be the set of solutions where $y_i > u_i$. Substituting $z_i = y_i - (u_i + 1) \ge 0$, the count $|A_i|$ becomes the unrestricted count with a reduced sum.
3. **Intersections:** $|A_i \cap A_j|$ substitutes both $z_i = y_i - (u_i + 1)$ and $z_j = y_j - (u_j + 1)$. If the remaining sum is negative, there are 0 solutions.

Formula:

$$\text{answer} = \sum_{S \subseteq \{1,\ldots,r\}} (-1)^{|S|} \binom{n - \sum_{i \in S}(u_i + 1) + r - 1}{r - 1},$$

where terms with a negative top entry in the binomial coefficient are 0.

Problem Statement

Find the number of integer solutions to

$$x_1 + x_2 + x_3 = 40, \qquad 4 \le x_1 \le 15, \quad 9 \le x_2 \le 18, \quad 5 \le x_3 \le 16.$$

Solution

Step 1: Shift variables to remove lower bounds.

Set $y_1 = x_1 - 4,\ y_2 = x_2 - 9,\ y_3 = x_3 - 5$. Then:

$$y_1 + y_2 + y_3 = 40 - 4 - 9 - 5 = 22,$$

with bounds $0 \le y_1 \le 11,\ 0 \le y_2 \le 9,\ 0 \le y_3 \le 11$.

Step 2: Count without upper bounds.

Without upper bounds, the number of non-negative integer solutions to $y_1 + y_2 + y_3 = 22$ is:

$$\binom{22+2}{2} = \binom{24}{2} = 276.$$

Step 3: Define violation sets and apply inclusion-exclusion.

Let A_i be the set of solutions where y_i exceeds its upper bound:

$$A_1 : \ y_1 \ge 12. \quad \text{Set } z_1 = y_1 - 12 \ge 0;\ z_1 + y_2 + y_3 = 10. \qquad |A_1| = \binom{12}{2} = 66.$$

$$A_2 : \ y_2 \ge 10. \quad \text{Set } z_2 = y_2 - 10 \ge 0;\ y_1 + z_2 + y_3 = 12. \qquad |A_2| = \binom{14}{2} = 91.$$

$$A_3 : \ y_3 \ge 12. \quad \text{Set } z_3 = y_3 - 12 \ge 0;\ y_1 + y_2 + z_3 = 10. \qquad |A_3| = \binom{12}{2} = 66.$$

Step 4: Pairwise intersections.

$A_1 \cap A_2$: $y_1 \geq 12,\ y_2 \geq 10$. Remaining sum: $22 - 12 - 10 = 0$. $\quad |A_1 \cap A_2| = \binom{2}{2} = 1$.

$A_1 \cap A_3$: $y_1 \geq 12,\ y_3 \geq 12$. Remaining sum: $22 - 12 - 12 = -2 < 0$. $\quad |A_1 \cap A_3| = 0$.

$A_2 \cap A_3$: $y_2 \geq 10,\ y_3 \geq 12$. Remaining sum: $22 - 10 - 12 = 0$. $\quad |A_2 \cap A_3| = \binom{2}{2} = 1$.

Step 5: Triple intersection.

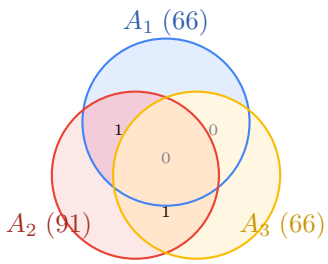
$$A_1 \cap A_2 \cap A_3 :\ y_1 \geq 12,\ y_2 \geq 10,\ y_3 \geq 12. \quad \text{Sum } \geq 34 > 22. \quad |A_1 \cap A_2 \cap A_3| = 0.$$

Step 6: Combine by inclusion-exclusion.

$$\begin{aligned} |A_1 \cup A_2 \cup A_3| &= (66 + 91 + 66) - (1 + 0 + 1) + 0 \\ &= 223 - 2 = 221. \end{aligned}$$

Final count:

$$276 - 221 = 55.$$



$|A_1 \cup A_2 \cup A_3| = 221$

Answer: $276 - 221 = 55$

The number of integer solutions is $\boxed{55}$.

Practice 8 — Recurrence Relations

Solutions to Selected Problems

Problems 2, 6

Discrete Mathematics — QYSU

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1 Problem 2 — Solving a Second-Order Linear Recurrence **2**

2 Problem 6 — Binary Strings with Two Consecutive Zeros **5**

1 Problem 2 — Solving a Second-Order Linear Recurrence

Theory Recap: Second-Order Linear Recurrences with Constant Coefficients

A **second-order linear recurrence with constant coefficients** has the form

$$a_n = c_1\, a_{n-1} + c_2\, a_{n-2}, \qquad n \geq 2,$$

where c_1, c_2 are constants and a_0, a_1 (or a_1, a_2) are given initial values.

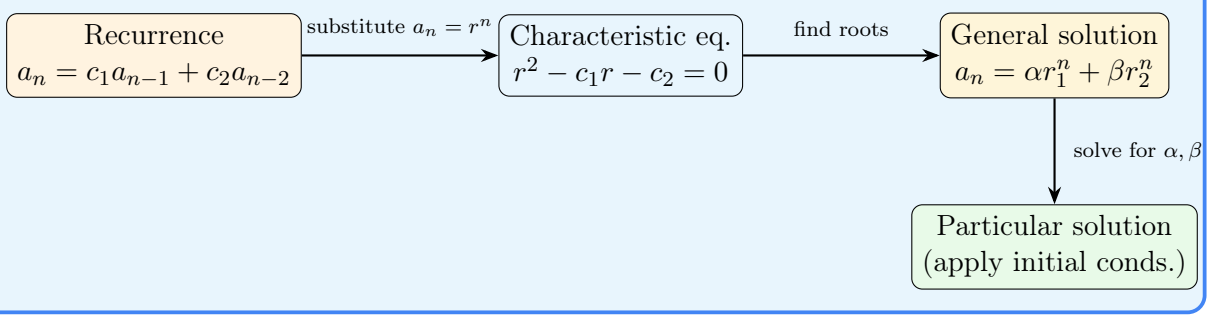
The Characteristic Equation Method. Substitute the trial solution $a_n = r^n$ into the recurrence to obtain the **characteristic equation**:

$$r^2 - c_1\, r - c_2 = 0.$$

The form of the general solution depends on the roots r_1, r_2:

Case	Roots	General Solution
Distinct real roots	$r_1 \neq r_2$, both real	$a_n = \alpha\, r_1^n + \beta\, r_2^n$
Repeated real root	$r_1 = r_2 = r$	$a_n = (\alpha + \beta\, n)\, r^n$
Complex conjugate roots	$r_{1,2} = \rho\, e^{\pm i\theta}$	$a_n = \rho^n \big(\alpha \cos n\theta + \beta \sin n\theta\big)$

The constants α and β are determined by substituting the initial conditions into the general solution and solving the resulting 2×2 linear system.



Problem Statement

Solve the linear recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2}, \qquad n > 2,$$

with initial conditions $a_1 = 2$ and $a_2 = 10$.

Solution

Step 1: Write the characteristic equation.

The recurrence $a_n - 5a_{n-1} + 6a_{n-2} = 0$ has characteristic equation

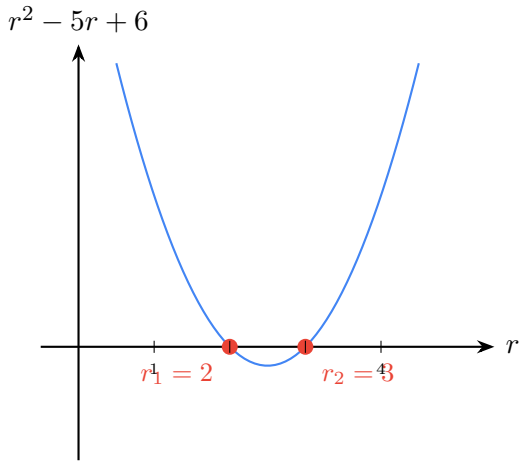
$$r^2 - 5r + 6 = 0.$$

Step 2: Find the roots.

Factoring:

$$r^2 - 5r + 6 = (r-2)(r-3) = 0 \qquad \Longrightarrow \qquad r_1 = 2, \quad r_2 = 3.$$

The roots are distinct and real, so we are in **Case 1**.



Step 3: Write the general solution.

Since $r_1 = 2$ and $r_2 = 3$ are distinct real roots:

$$a_n = \alpha \cdot 2^n + \beta \cdot 3^n.$$

Step 4: Apply initial conditions.

Substituting $n = 1$ and $n = 2$:

$$\begin{cases} a_1 = 2\alpha + 3\beta = 2 \\ a_2 = 4\alpha + 9\beta = 10 \end{cases}$$

From the first equation:

$$\alpha = \frac{2 - 3\beta}{2} = 1 - \frac{3\beta}{2}.$$

Substituting into the second equation:

$$4\left(1 - \frac{3\beta}{2}\right) + 9\beta = 10 \quad \Longrightarrow \quad 4 - 6\beta + 9\beta = 10 \quad \Longrightarrow \quad 3\beta = 6 \quad \Longrightarrow \quad \beta = 2.$$

Then $\alpha = 1 - \dfrac{3 \cdot 2}{2} = 1 - 3 = -2$.

Step 5: Write the particular solution.

$$a_n = -2 \cdot 2^n + 2 \cdot 3^n = 2\big(3^n - 2^n\big).$$

Step 6: Verify.

n	$2(3^n - 2^n)$	Expected	Match
1	$2(3-2) = 2$	$a_1 = 2$	✓
2	$2(9-4) = 10$	$a_2 = 10$	✓
3	$2(27-8) = 38$	$5(10) - 6(2) = 38$	✓
4	$2(81-16) = 130$	$5(38) - 6(10) = 130$	✓

$$\boxed{a_n = 2\left(3^n - 2^n\right)}$$

2 Problem 6 — Binary Strings with Two Consecutive Zeros

Theory Recap: Complementary Counting and Recurrences for Constrained Strings

Complementary counting. When counting objects that satisfy a property P, it is sometimes easier to count those that do *not* satisfy P:

$$|P| = |\text{total}| - |\text{not } P|.$$

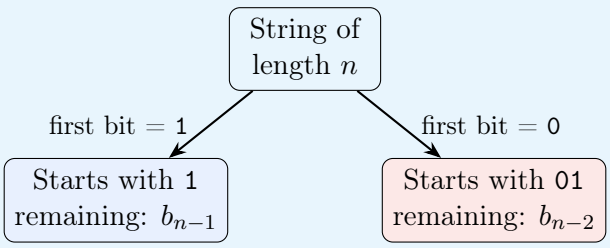
Setting up recurrences for constrained strings. To count binary strings of length n satisfying (or avoiding) a pattern:

1. Define a sequence: let b_n be the count of valid strings of length n.
2. **Condition on the first (or last) character.** Each choice constrains what can follow, yielding a recurrence.
3. Solve using base cases.

Example pattern: avoiding "00". For a string of length n with no two consecutive 0s:

- If it starts with `1`: the remaining $n-1$ characters form any valid string $\longrightarrow b_{n-1}$ choices.
- If it starts with `0`: the next character *must* be `1` (to avoid "00"), and the remaining $n-2$ characters form any valid string $\longrightarrow b_{n-2}$ choices.

This gives the Fibonacci-like recurrence $b_n = b_{n-1} + b_{n-2}$.



Problem Statement

How many binary strings of length 10 contain two consecutive zeros?

Solution

We use **complementary counting**.

Step 1: Count all binary strings of length 10.

Each position has 2 choices, so the total is $2^{10} = 1024$.

Step 2: Count binary strings of length 10 with *no* two consecutive zeros.

Let b_n denote the number of binary strings of length n with no two consecutive 0s.

Deriving the recurrence. Consider a valid string of length n:

- If it starts with `1`: the remaining $n-1$ characters can be any valid string of length

$n-1$, giving b_{n-1} choices.

- If it starts with `0`: the next character must be `1` (otherwise we have "00"), and the remaining $n-2$ characters can be any valid string of length $n-2$, giving b_{n-2} choices.

Therefore:

$$b_n = b_{n-1} + b_{n-2}, \qquad n \geq 3.$$

Base cases.

- $b_1 = 2$: the strings are `0` and `1`.
- $b_2 = 3$: the valid strings are `01`, `10`, `11` (excluding `00`).

Step 3: Compute b_n for $n = 3, 4, \ldots, 10$.

n	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
b_n	2	3	5	8	13	21	34	55	89	144

$b_9 = b_8 + b_7 = 55 + 34 = 89$

Detailed computation:

$$\begin{aligned}
b_3 &= b_2 + b_1 = 3 + 2 &&= 5 \\
b_4 &= b_3 + b_2 = 5 + 3 &&= 8 \\
b_5 &= b_4 + b_3 = 8 + 5 &&= 13 \\
b_6 &= b_5 + b_4 = 13 + 8 &&= 21 \\
b_7 &= b_6 + b_5 = 21 + 13 &&= 34 \\
b_8 &= b_7 + b_6 = 34 + 21 &&= 55 \\
b_9 &= b_8 + b_7 = 55 + 34 &&= 89 \\
b_{10} &= b_9 + b_8 = 89 + 55 &&= 144
\end{aligned}$$

Observation. The values $b_n = 2, 3, 5, 8, 13, 21, 34, 55, 89, 144$ are the Fibonacci numbers $F_3, F_4, \ldots, F_{12}$. In general, $b_n = F_{n+2}$ where $F_1 = 1, F_2 = 1, F_3 = 2, \ldots$

Step 4: Apply complementary counting.

$$\text{Strings with "00"} = \text{Total} - \text{Strings without "00"} = 1024 - 144 = 880.$$

Verification for small cases. For $n = 2$: total $= 4$, strings without "00" $= 3$ (`01`, `10`, `11`), so strings with "00" $= 1$ (namely `00`). ✓

For $n = 3$: total $= 8$, $b_3 = 5$ (`010`, `011`, `101`, `110`, `111`), so strings with "00" $= 3$ (`001`, `100`, `000`). ✓

The number of binary strings of length 10 that contain two consecutive zeros is $\boxed{880}$.